# Report of Geotechnical Exploration (Revision 1)
# FSR 221 Peavine Sheeds Creek
# Conasauga, Tennessee
# S&ME Project No. 23810276

PREPARED FOR:

**WSP USA, Inc.**
**13530 Dulles Technology Drive, Suite 300**
**Herndon, Virginia 20171**

PREPARED BY:


**S&ME, Inc.**
**4291 Highway 58**
**Chattanooga, Tennessee 37416**


**February 24, 2026**

February 24, 2026

WSP USA, Inc.
13530 Dulles Technology Drive, Suite 300
Herndon, Virginia 20171

Attention: Mr. Matthew Martin

Reference: **Report of Geotechnical Exploration (Revision 1)**
**FSR 221 Peavine Sheeds Creek Road**
Forest Service Road 221, Conasauga, Tennessee
S&ME Project No. 23810276

Dear Mr. Martin:

This report presents the results of the geotechnical exploration for the FSR221 Peavine Sheeds Creek Road site in Conasauga, Tennessee. Our services were provided in general accordance with S&ME Proposal No. 23810276, dated December 3, 2023, as authorized by the Professional Services On-Call/Task Order Subcontract between WSP USA, Inc. and S&ME dated November 14, 2023.

This report describes our understanding of the project, presents the results of the field exploration and laboratory testing, and discusses our conclusions and recommendations. S&ME appreciates this opportunity to be of service to you. Please call if you have questions concerning this report or any of our services.

Sincerely,

**S&ME, Inc.**

David Grass, PE
Project Engineer
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Christopher Brown, PG PMP
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Senior reviewed by: Daniel R. Boles, PE (formerly of S&ME) / Revision 1 reviewed by Christopher Brown, PG PMP

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# 1.0 Introduction

The primary purpose of our services was to explore the subsurface conditions at the site, provide laboratory testing, and provide a geotechnical engineering report including a retaining wall design. Plans for the retaining wall design will be submitted separately from this report. This report describes our understanding of the project, presents the results of the field exploration and laboratory testing, and discusses our conclusions and recommendations relative to the considerations discussed in this report. This report provides the following:

- A summary of the project and provided information;
- A summary of current site conditions, topography, and area geology;
- A summary of the field exploration methods;
- A summary of the subsurface conditions encountered in the test borings;
- A summary of the laboratory test methods and results;
- Recommendations for site preparation, including subgrade preparation, excavation, structural fill placement, and groundwater control;
- Recommendations for foundation design and construction for each structure;
- Results of slope stability analysis;
- Seismic hazard analysis of the sites following AASHTO LRFD Section 3.10; and
- An Appendix with Site Location Plans, Boring Location Plans, individual boring logs for each test boring, site photos, and individual laboratory test reports.

# 2.0 Project Description

Initial project information was provided through emails from and conversations with Mr. Matthew Martin of WSP. Appended to the emails were the following document:

- *Design Scoping Report TN ERFO FS CHRK804 202-2 (1),* by the Federal Highway Administration (FHWA), May 2023.
- *Section B Supplies or Services and Prices/Costs,* by FHWA (undated).
- *Section C Statement of Work,* by FHWA (undated).
- DGN files of the existing condition surveys for each site.

Based on the provided information, we understand there has been storm damage impacting Forest Service Road 221 (Peavine Sheeds Creek Road). Amongst the planned repairs are the replacement of a culvert at Mile Post 6.229, the construction of a retaining wall to repair a slope failure at Mile Post 7.84, and the construction of an aquatic organism passage (AOP) at Mile Post 19.302. Site Location Plans, Figures 1 through 3, are included in the Appendix showing the general location of each of these sites.

Based on provided information, we understand the proposed box culvert located at Mile Post 6.229 will be a precast concrete structure bearing on shallow foundations. Based on conversations with the project team, we understand the preference for the retaining wall constructed at the slope failure located at Mile Post 7.84 is a mechanically stabilized earth (MSE) wall type. Lastly, we understand the proposed AOP located at Mile 19.302 will consist of a bottomless, half-dome corrugated metal pipe supported on shallow concrete foundations.

The project information detailed in this report should be reviewed and confirmed by the appropriate team members. Modifications to our conclusions and recommendations may be required if the actual conditions vary from the project information and assumptions described herein.

# 3.0 Exploratory and Testing Procedures

## 3.1 Soil Test Borings

A member of S&ME's professional staff made a site reconnaissance to observe site features and to mark the test boring locations. The locations of the borings, labeled B-01 through B-04, were selected at each site based on the existing site features, obstructions, and drill rig access. Members of the client's survey subcontractor, Cannon and Cannon, later surveyed the locations of these borings. The approximate boring locations are shown on the Boring Location Plans (Figures No. 4 through 6) in the Appendix. Ground surface elevations at the boring locations were estimated to the nearest foot based on the approximate locations and interpolation between the contours shown on the provided site plan. These elevations should be considered approximate.

Due to the rocky nature of the overburden soils, we were not able to advance our soil test borings using hollow stem augers as originally proposed. Therefore, the soil test borings for this exploration were made using a casing advancer in general accordance with ASTM D5872, *Standard Practice for Use of Casing Advancement Drilling Methods for Geoenvironmental Exploration*. Continuous soil sampling was performed in general accordance with ASTM D1586, *Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils* until auger refusal was reached for each boring. During standard penetration testing, the sampler was first seated 6 inches and then driven an additional foot with blows of a 140-pound hammer falling 30 inches. The number of hammer blows required to drive the sampler the final foot was recorded and is designated the *standard penetration resistance* (SPT) or *N-value* with units of blows per foot (bpf).

In addition to split barrel sampling, one relatively undisturbed Shelby tube sample was obtained from select boring locations and depths. We were unable to obtain more Shelby Tubes due to the granular nature of the materials. The borings were then backfilled with bentonite grout before leaving the site.

## 3.2 Laboratory Testing

The samples obtained during the field exploration were transported to our laboratory and visually classified by members of our professional staff in general accordance with ASTM D2488 *Standard Practice for Description and Identification of Soils (Visual-Manual Procedure)*. The purposes of this review were to check the field descriptions, visually estimate the relative percentages of the soils' constituents (sand, clay, etc.), and evaluate the soil's origin. The stratification lines shown on the appended Boring Logs represent the approximate boundaries between soil types, but the transitions may be more gradual than shown. Selected soil samples were subjected to moisture content (ASTM D2216), Atterberg limits testing (ASTM D4318), grain size distribution (ASTM D1140), unconfined compression strength of rock (ASTM D2166), and corrosion testing including pH, sulfate, and chloride concentrations. We were unable to perform triaxial testing because testable Shelby tube samples could not be obtained in the soil matrix.

# 4.0 Site, Geologic, and Subsurface Conditions

## 4.1 Site Conditions

Each of the three sites are located along Forest Service Road FSR 221, also known as Peavine Sheeds Creek Road, within the southern Cherokee National Forest in eastern Tennessee. The sites are located at mile posts 6.229, 7.84, and 19.302. The site at Mile Post 6.229 is generally characterized by a small creek crossing currently accommodated by a metal culvert. The site at Mile Post 7.84 has a failing slope above the adjacent Graham Creek. The site at Mile Post 19.302 comprises a creek crossing above the North Prong of Brown Camp Branch. Each site is located along a gravel, or asphalt-paved forest service road surrounded by mature trees and is adjacent to a stream.

## 4.2 Geologic Conditions

The project site lies within the Blue Ridge Physiographic Province (also known as the Unaka Physiographic Province). The Blue Ridge, located along the eastern border of Tennessee, is a deeply dissected mountainous area of numerous steep mountain ridges (with elevations up to about 6,600 feet), basins, and trench valleys that intersect at all angles which give the area its rugged mountain character. This province is underlain by complexly folded and faulted igneous and metamorphic rock with some sedimentary rock. The Blue Ridge contains the highest elevations and the most rugged topography in the Appalachian Mountain system of eastern North America.

Based on our review of the Geologic Map of East Tennessee, Plate 13, Cleveland Quadrangle, dated 1953, each site is underlain by bedrock of the Sandsuck Formation of the Walden Creek Group. This formation consists of greenish-gray siltstone, silty shale, and fine-grained sandstone with thick lenticular beds of sandstone and quartz-pebble conglomerate that occurs in the middle part of the formation. A copy of these maps is included in Appendix I.

## 4.3 Subsurface Conditions

The borings logs included in the Appendix should be reviewed for specific information at the individual boring locations. The depths and thicknesses of the subsurface strata indicated on the boring logs are generalized and the transition between materials may be more gradual than indicated. Information on actual subsurface conditions exists only at the specific boring locations and is relevant to the time the exploration was performed. Variations may occur and should be expected between and away from the boring locations. The stratification lines were used for our analytical purposes and, unless specifically stated otherwise, should not be used as the basis for design or construction cost estimates.

#### 4.3.1 *Surface Materials*

Topsoil was encountered in borings B-01, B-02, and B-04 to a depth of about 1 inch. Asphalt was encountered in boring B-03 at a depth of about 2 inches.

#### 4.3.2 *Residual Soils*

Residual soils were encountered in each of the soil test borings below the surface materials to auger refusal depths at about 4 inches to 10 feet. Residual soil forms from the in-place weathering of the underlying bedrock. The residual soils encountered at the site were typically composed of orange brown to gray gravels and sands with varying amounts of silt or clay. N-values in the residuum ranged from 8 blows per foot to over 50 blows per 6-inch increment in cohesive soils, indicating a stiff to hard consistency. N-values in residual granular soils ranged from 4 to over 50 blows per 6-inch increment, indicating very loose to very dense relative densities.

#### 4.3.3 *Auger Refusal Materials*

Auger refusal materials were encountered in each of the borings at depths ranging from about 4 inches to 10 feet below the existing ground surface. Refusal occurs when materials are encountered that cannot be penetrated by the drilling equipment and is normally indicative of a very hard or very dense material, such as a boulder, a rock lens or pinnacle, or the upper surface of bedrock. Auger refusal discussed herein is based on conditions impenetrable to the drilling equipment for this project. Auger refusal conditions with this particular equipment do not necessarily indicate conditions impenetrable to other equipment.

Rock was cored in each of the borings. The rock core samples collected were logged in the field. The recovery and rock quality designation (RQD) were determined prior to storing the samples. The rock core recovery is a measure of the length of core drilled to that recovered expressed as a percentage. The RQD is a measure of the rock quality based on the percentage of the rock core run containing pieces greater than 4 inches in length. The rock core recovery ranged from 85 to 100 percent. The RQD ranged from 60 to 100 percent. The rock cored typically consisted of gray and dark gray, moderately hard to very hard sandstones and shales.

## 5.0 Laboratory Testing

### 5.1 Laboratory Test Results Summary

The results of the laboratory testing for this project are summarized in the following table with the exception of grain size distribution and corrosion testing. Laboratory test reports are included in the Appendix.

**Test Results Summary**

| Boring | Sample Type | Depth (ft) | Natural Moisture Content (%) | Percent Finer than US No 200 Sieve (%) | Liquid Limit | Plasticity Index | Unconfined Compressive Strength (psi) |
|---|---|---|---|---|---|---|---|
| B-01 | Split Spoon | 0-2 | 8.9 | - | - | - | - |
| B-02 | Split Spoon | 0-2 | - | 43.2 | 38 | 13 | - |
| | Split Spoon | 2-4 | - | 9.4 | 35 | 12 | - |
| | Split Spoon | 4-6 | 5.2 | - | - | - | - |
| | Split Spoon | 6-8 | 1.6 | - | - | - | - |
| | Rock Core | 9.1-9.5 | - | - | - | - | 35,500 |
| B-03 | Split Spoon | 0-2 | - | 29.4 | 30 | 10 | - |
| | Split Spoon | 2-4 | - | 15.3 | 27 | 8 | - |
| | Rock Core | 9.7-10.1 | - | - | - | - | 20,482 |
| B-04 | Split Spoon | 0-2 | 26.8 | 63.4 | 39 | 10 | - |
| | Split Spoon | 4-6 | 33.3 | 56.1 | 47 | 10 | - |
| | Split Spoon | 6-8 | 34.0 | - | - | - | - |

"-" Test not performed

### 5.2 Soil Corrosivity

Split-spoon samples obtained from each boring were selected for soil corrosivity testing by WSP. The testing included the determination of pH (ASTM G51), resistivity (ASTM G 187), chloride ion concentration (ASTM D 4327), and sulfate ion concentration (ASTM D4327). The results of these tests are used to evaluate the aggressiveness (corrosivity) of the subsurface environment with respect to foundation materials such as concrete and steel. The sample recovered from B-01 did not have a sufficient mass of material and was not tested for resistivity.

## 6.0 Conclusions and Recommendations

### 6.1 Earthwork

#### *6.1.1 Stripping*

Surface materials, including topsoil, gravel, and asphalt, should be stripped from the construction area. Topsoil was encountered to a depth of about one inch in borings B-01, B-03, and B-04 which were performed near the shoulders of the existing gravel roads. However, we expect the topsoil thicknesses may be greater in unexplored areas of the site particularly near drainage features. The amount of topsoil stripped prior to construction may be greater than the depths identified in our test borings depending on limitations of the contractor's equipment.

Approximately 2 inches of asphalt was encountered in boring B-02. We also observed gravel paving at Mile Posts 6.229, 7.84, and 19.302. We expect this gravel paving was several inches thick. However, greater depths of paving materials may be present in unexplored areas.

#### 6.1.2 *Subgrade Evaluation*

After completion of stripping in areas to receive fill, and once grade is achieved in cut areas, we recommend a member of our geotechnical engineering staff evaluate the stability of the exposed subgrade soils. Subgrade evaluations may include hand probing with a steel probe rod, hand auger borings, test pit excavations or proofrolling with a loaded dump truck or other heavy equipment. In general, unstable materials should be undercut to stable soils and backfilled with compacted structural fill unless otherwise directed by the geotechnical engineer or his representative.

Subgrade repair can be expected to be more extensive if grading operations are performed during wet periods of the year. Once areas that need remediation have been repaired, the site may be brought to grade with structural fill. Depending on prevailing weather conditions and the speed of contractor activities, subgrade evaluations may be required on multiple occasions.

#### 6.1.3 *Difficult Excavation*

Based on the boring data obtained during the exploration, we expect material requiring difficult excavation techniques will be encountered during foundation construction at the proposed box culvert and possibly at the AOP structure. Auger refusal was encountered at a depth of about ½ foot in boring B-01 (Miles Post 6.229) and a depth of about 10 feet in boring B-04 (Mile Post 19.302). In confined excavations such as foundations, removal of weathered rock typically requires the use of large backhoes and/or pneumatic spades. The difficulty of excavation will depend on the composition of the rock, the location and orientation of discontinuities and bedding, and the skill of the equipment operator.

#### 6.1.4 *Earth Material Utilization and Fill Placement*

After subgrade evaluation/preparation and completion of any field-recommended remedial actions, areas to receive fill may be brought to design subgrade levels with structural fill. Structural fill is defined as inorganic natural soil with maximum particle sizes of about 4 inches and a Plasticity Index of 30 or less. Structural fill should be placed in relatively thin (4- to 8-inch) layers and compacted to at least 98 percent of the soil's maximum dry density as determined by the standard Proctor compaction test (ASTM D698).

#### 6.1.5 *Fill Density Testing*

In-place density testing must be performed as a check the previously recommended compaction criteria are achieved. We recommend density testing by a technician working under the direction of our project engineer.

#### 6.1.6 *Groundwater*

Based on the test boring results, we do not expect that groundwater will present significant site development problems. However, perched groundwater conditions may occur at this site, particularly during periods of heavy rain. The contractor should provide adequate dewatering to maintain the groundwater level, if encountered, and surface water below the bottom of excavations, except when the excavation is in rock. Pumping from sumps constructed outside the excavation should be effective in dewatering the soils encountered in our exploration. However, the dewatering methods selected should take into consideration the presence of gravel layers and pockets that will concentrate the groundwater flow in discreet zones. Water from the pumps should be discharged beyond the construction boundaries to limit its effect on construction activities.

## 6.2 Final Subgrade Preparation

Roadway subgrades are often disturbed between completion of site grading and final roadway construction due to weather, foundation installation, and other construction activities. For this reason, it is important the subgrades be evaluated just before final roadway grading. At that time, as much of the subgrade as practical should be proofrolled with a loaded dump truck in the presence of our representative. Any areas that deflect significantly under the proofrolling load, or which are otherwise assessed to be soft or unstable, should be undercut to firm materials. The undercut areas should be backfilled with compacted soil or crushed stone. Our representative can provide recommendations for repairing unstable areas that are observed.

## 6.3 Foundations

### *6.3.1 Box Culvert – Master Pin 77 / Mile Post 6.229*

Based on the results of our soil test borings, we expect intact rock will be encountered at or above the proposed foundation bearing elevation of the proposed box culvert. We recommend the foundations be supported entirely on rock. If residual soils are encountered in areas at this proposed bearing elevation, this material should be undercut to expose intact bedrock. Once the excavations are completed and relatively free of loose debris, they should be backfilled with flowable fill or lean concrete to the design foundation bearing elevation. Foundation concrete may then be placed neat to the excavation.

Based on our analysis, foundations prepared in accordance with the recommendations of this report and bearing on bedrock, or on flowable fill, may be designed using an allowable bearing pressure of 3,000 pounds per square foot.

### *6.3.2 Retaining Wall – Master Pin 92 / Mile Post 7.84*

The proposed retaining wall at Mile Post 7.84 will be a MSE wall. The proposed wall will include a block facing supported on a unreinforced cast-in-place concrete leveling pad and geogrid reinforced zone. Recommendations for MSE wall construction will be provided in the *Mechanically Stabilized Earth Retaining Wall Design – Peavine Sheeds Creek Road Slope Stabilization* which will be issued under separate cover. A subsurface profile (Figure No. 7) at the proposed wall location is included in Appendix II.

### *6.3.3 Aquatic Organism Passage – Master Pin 179 / Mile Post 19.302*

Based on the results of our soil test borings, we expect residual soils will be encountered at proposed foundation bearing elevations for the aquatic organism passage site. Based on our analysis, shallow spread footings bearing on compacted soil fill or medium dense or higher relative density or stiff or higher consistency residual soil may be designed using an allowable soil bearing pressure of 1,500 pounds per square foot. If a higher bearing pressure is required, the foundations may be undercut to bedrock and backfilled with lean concrete in accordance with the recommendations in Section 6.3.1. An allowable bearing pressure of up to 3,000 psf would be available for foundations bearing on rock.

### *6.3.4 General Foundation Recommendations*

Foundation excavations should be backfilled with concrete the same day they are excavated. Footing concrete should be placed directly against the excavation so water cannot collect behind forms before backfilling. If soils exposed in the foundation excavations experience moisture variations prior to concrete placement, the affected bearing materials should be undercut as recommended by our geotechnical engineer. A 2- to 3-inch thick mud-mat of lean concrete may be used to protect the exposed support materials if the foundations cannot be backfilled with concrete the same day they are excavated. Although computed footing dimensions may be less, we recommend that continuous footings be a minimum of 18 inches wide and isolated spread footings be a minimum of 36 inches wide. Foundations should bear at a minimum of 18 inches below subgrade to protect against frost penetration and to provide adequate confinement.

The recommendations in this report are contingent on S&ME observing and evaluating the foundation excavations prior to placing concrete. Foundation subgrade observations should be performed by the geotechnical engineer, or his qualified representative, to confirm the recommendations provided in this report are consistent with the site conditions encountered. A Dynamic Cone Penetrometer (DCP) should be utilized to provide information that is compared to the data obtained in the geotechnical report. If unacceptable materials are encountered, the material should be excavated to stiff or higher consistency soils or remediated as recommended by the geotechnical engineer.

The recommendations in this report are contingent on S&ME observing and evaluating the foundation excavations prior to placing flowable fill or concrete. Foundation subgrade observations should be performed by the geotechnical engineer, or his qualified representative, in order to confirm the recommendations provided in this report are consistent with the site conditions encountered.

## 6.4 Seismic Considerations (AASHTO 2020)

The soil profile present at each site has been evaluated for a seismic design classification utilizing standard penetration resistance (N-value) information in general accordance with AASHTO LRFD Bridge Design Specifications, 2020 Section 3.10.3. Based on the results of our exploration and the geology of the area, we recommend a Site Class of C be used for the design of each of the proposed structures. The AASHTO manual contains a provision for assessing the use of site-specific values, provided a site-specific assessment is conducted. S&ME can provide these services, if requested. For a site class of C for each structure, we determined the following design spectral acceleration parameters in accordance with AASHTO 3.10.2.1 provided in the table below.

### Spectral Acceleration Parameters

| Structure | Master Pin | PGA | $S_{DS}$ | $S_{D1}$ |
|---|---|---|---|---|
| Box Culvert | 77 | 0.148 | 0.321 | 0.117 |
| Wall | 92 | 0.147 | 0.320 | 0.117 |
| Aquatic Organism Passage | 179 | 0.146 | 0.319 | 0.117 |

## 6.5 Corrosivity

To evaluate the corrosion potential of onsite soils, we reviewed our corrosivity lab test results and compared them to *Table 5-15 Typical Numerical Corrosivity Scoring System* of the *FHWA-NHI-16-072 GEC 5 – Geotechnical Site Characterization* document. Corrosivity scores for each site are summarized in the following table. The lab tests are included in the Appendix.

### Corrosion Potential Score Summary

| Test Location | Total Corrosivity Score | Soil Corrosion Potential |
|---|---|---|
| B-01 | 7* | Moderate |
| B-02 | 2 | Unlikely |
| B-03 | 1 | Unlikely |
| B-04 | 9 | Aggressive |

*We were not able to obtain enough sample to perform resistivity testing on the samples obtained from boring B-01. The corrosivity score assumes a resistivity value of greater than 10,000 Ω-cm. Should resistivity of the soils at B-01 be less than 10,000 Ω-cm, this material would likely be considered to have an aggressive soil corrosion potential.

# 7.0 Follow-Up Services

Our services should not end with the submission of this geotechnical report. S&ME should be kept involved throughout the design and construction process to maintain continuity and to determine if our recommendations are properly interpreted and implemented. To achieve this, we should review project plans and specifications with the designers to see that our recommendations are fully incorporated and have not been misinterpreted. We also should be retained by the owner to observe the site preparation and foundation construction.

S&ME's familiarity with the site and foundation recommendations makes us a valuable part of your construction quality assurance team. S&ME recommends we be retained by the owner on a full-time basis to observe earthwork and foundation construction. Our personnel are uniquely qualified to recognize unanticipated ground conditions and can offer responsive remedial recommendations should these unanticipated conditions occur.

## 8.0 Limitations of Report

This report has been prepared in accordance with generally accepted geotechnical engineering practice for specific application to this project. The conclusions and recommendations contained in this report are based upon applicable standards of our practice in this geographic area at the time this report was prepared. No other representation or warranty is express or implied.

We relied on project information given to us or assumed by us based on similar projects in the East Tennessee area to develop our conclusions and recommendations. If project information described in this report is not accurate, or if it changes during project development, we should be notified of the changes so that we can modify our recommendations based on this additional information, if appropriate.

Our conclusions and recommendations are based on limited data from a field exploration program. Subsurface conditions can vary widely between explored areas. Some variations may not become evident until construction. If conditions are encountered which appear different than those described in our report, we should be notified. This report should not be construed to represent subsurface conditions for the entire site.

Unless specifically noted otherwise, our field exploration program did not include an assessment of regulatory compliance, environmental conditions or pollutants or presence of any biological materials (mold, fungi, bacteria). If there is a concern about these items, other studies should be performed. S&ME can provide a proposal and perform these services, if requested.

S&ME should be retained to review the final plans and specifications to check that earthwork, foundation, and other recommendations are properly interpreted and implemented. For additional information regarding the use and limitations of this report, please read the *Important Information about your Geotechnical Engineering Report* document located at the end of this report.

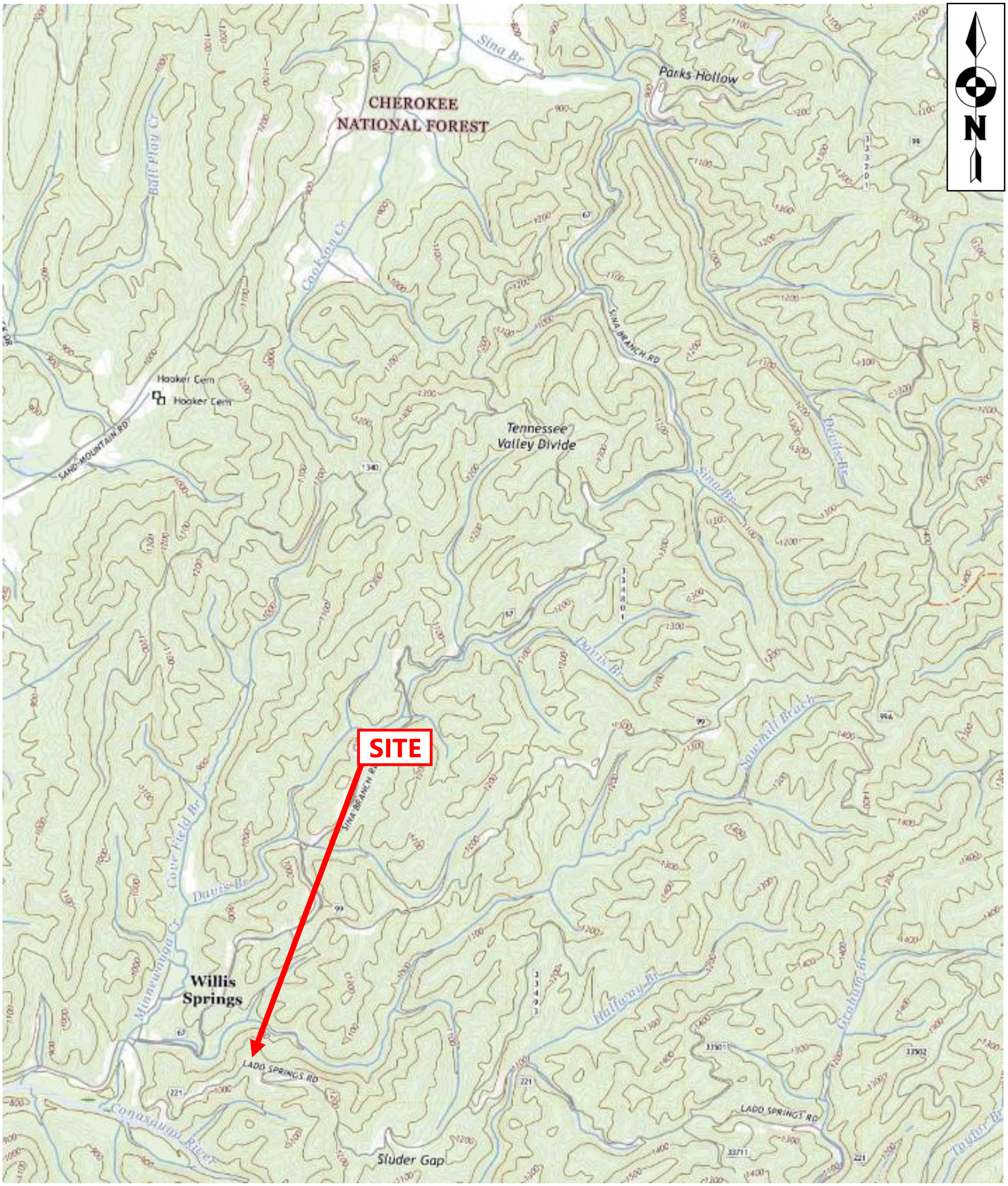
# Appendix

# Appendix I

Site Location Plans

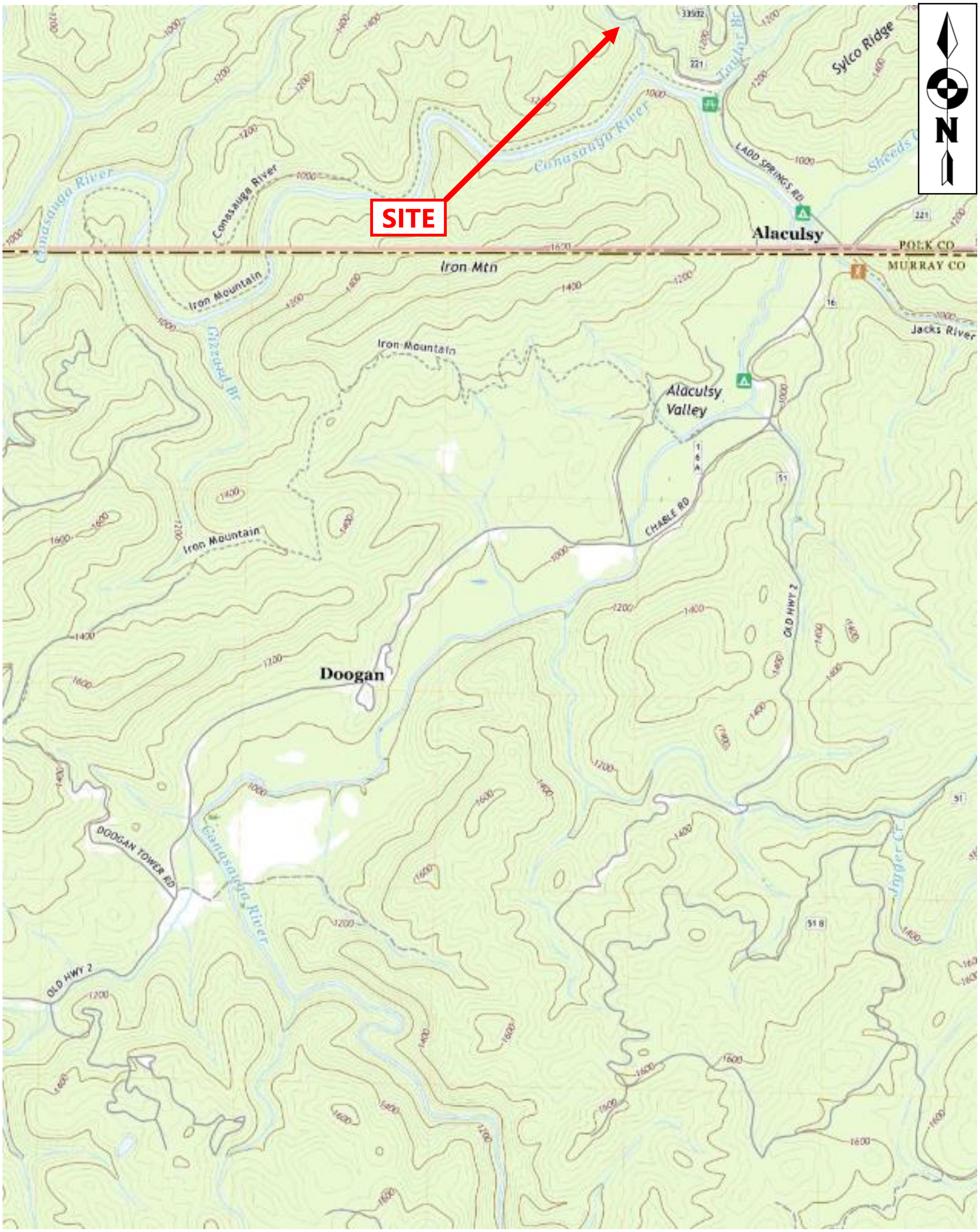
Boring Location Plans

Geologic Maps




DRAWING FOR ILLUSTRATION PURPOSES ONLY
SOURCE: USGS TOPOGRAPHICAL MAP, PARKSVILLE TN 2022

| | | | |
|---|---|---|---|
| **SITE LOCATION PLAN – MASTER PIN 77/MILE POST 6.229** | SCALE: NOT TO SCALE | DATE: 5/29/2024 | FIGURE NO. **1** |
| FSR 221 PEAVINE SHEEDS CREEK<br>CONASAUGA, TENNESSEE | PROJECT NUMBER 23810276 | | |




**DRAWING FOR ILLUSTRATION PURPOSES ONLY**
**SOURCE: USGS TOPOGRAPHICAL MAP, TENNGA GA 2022**

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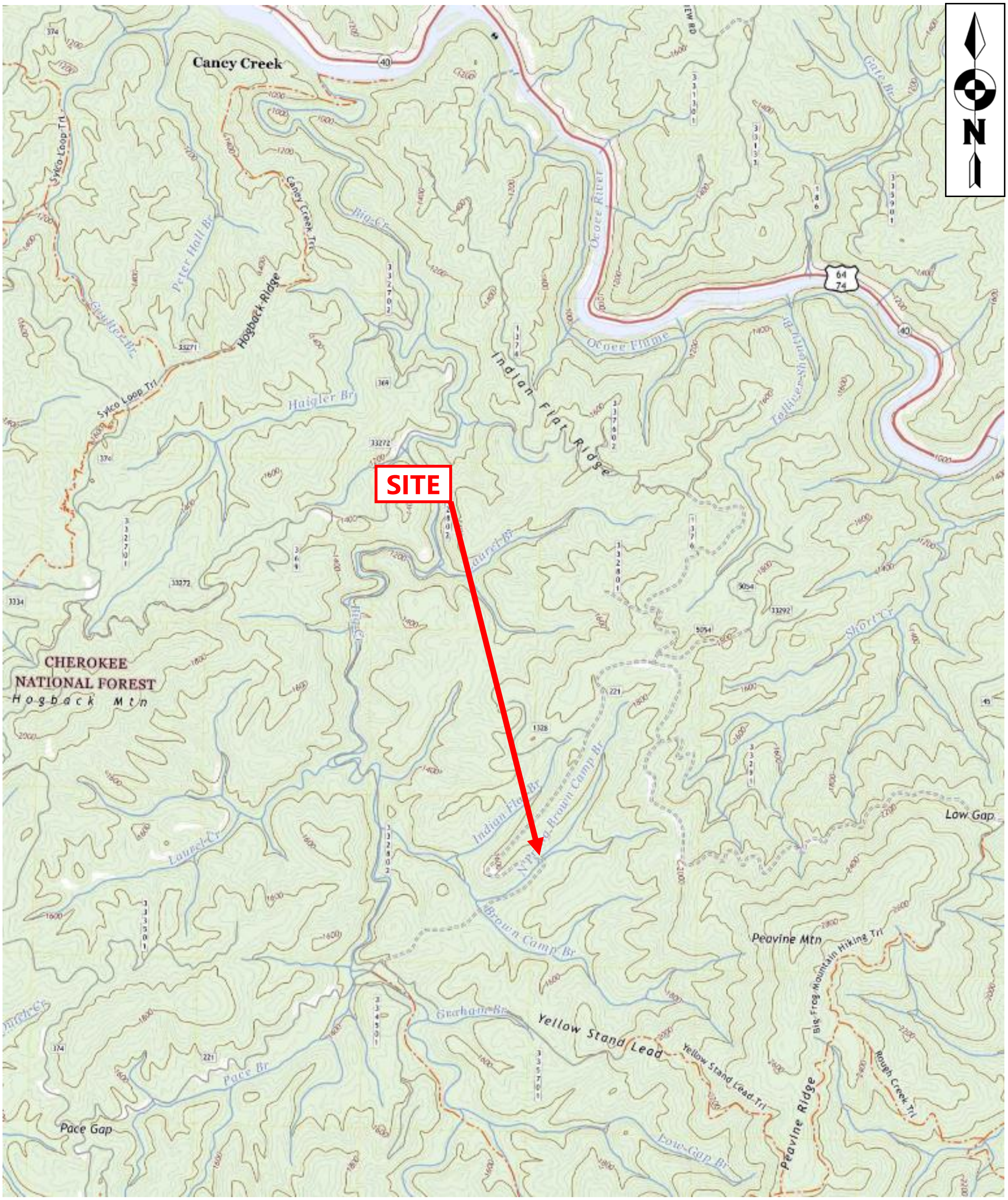
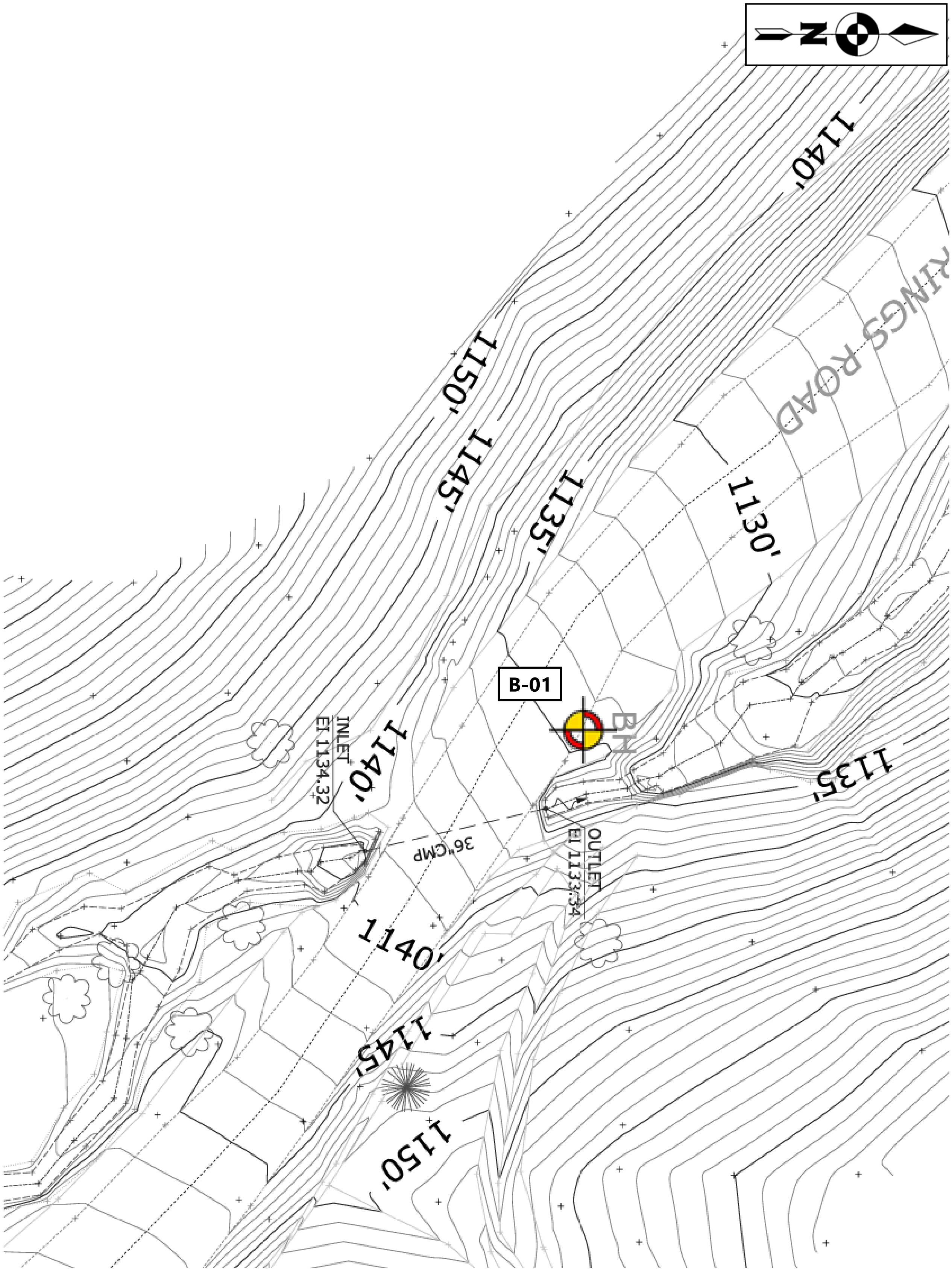
|---|---|---|---|
| S&ME | **SITE LOCATION PLAN – MASTER PIN 92/MILE POST 7.84** | SCALE: NOT TO SCALE | FIGURE NO. |
| | FSR 221 PEAVINE SHEEDS CREEK<br>CONASAUGA, TENNESSEE | DATE: 5/29/2024<br>PROJECT NUMBER 238103276 | **2** |




DRAWING FOR ILLUSTRATION PURPOSES ONLY
SOURCE: USGS TOPOGRAPHICAL MAP, CANEY CREEK TN 2022

| | | |
|---|---|---|
| **SITE LOCATION PLAN – MASTER PIN 179/MILE POST 19.302** | SCALE: NOT TO SCALE<br>DATE: 5/29/2024 | FIGURE NO. 3 |
| FSR 221 PEAVINE SHEEDS CREEK<br>CONASAUGA, TENNESSEE | PROJECT NUMBER<br>23810276 | |



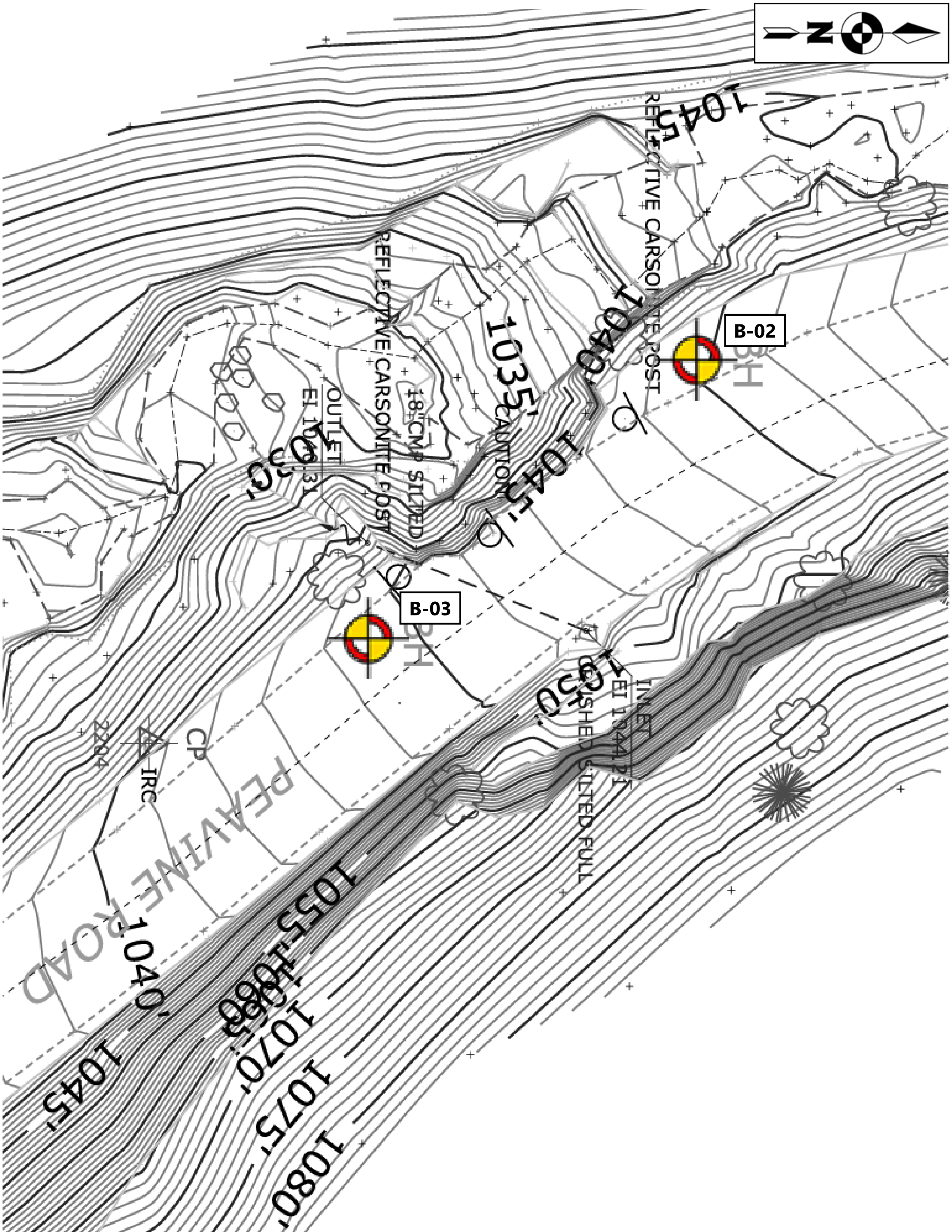

**NOTES:**
**IMAGERY PROVIDED BY WSP.**
**DRAWING FOR ILLUSTRATION PURPOSES ONLY**

**LEGEND:** **APPROXIMATE BORING LOCATION**

**BORING LOCATION PLAN – MASTER PIN 77/MILE POST 6.229**

FSR 221 PEAVINE SHEEDS CREEK
CONASAUGA, TENNESSEE

SCALE: NOT TO SCALE
DATE: 5/29/2024
PROJECT NUMBER 23810276
FIGURE NO. 4




**NOTES:**
**IMAGERY PROVIDED BY WSP.**
**DRAWING FOR ILLUSTRATION PURPOSES ONLY**

**LEGEND:** **APPROXIMATE BORING LOCATION**

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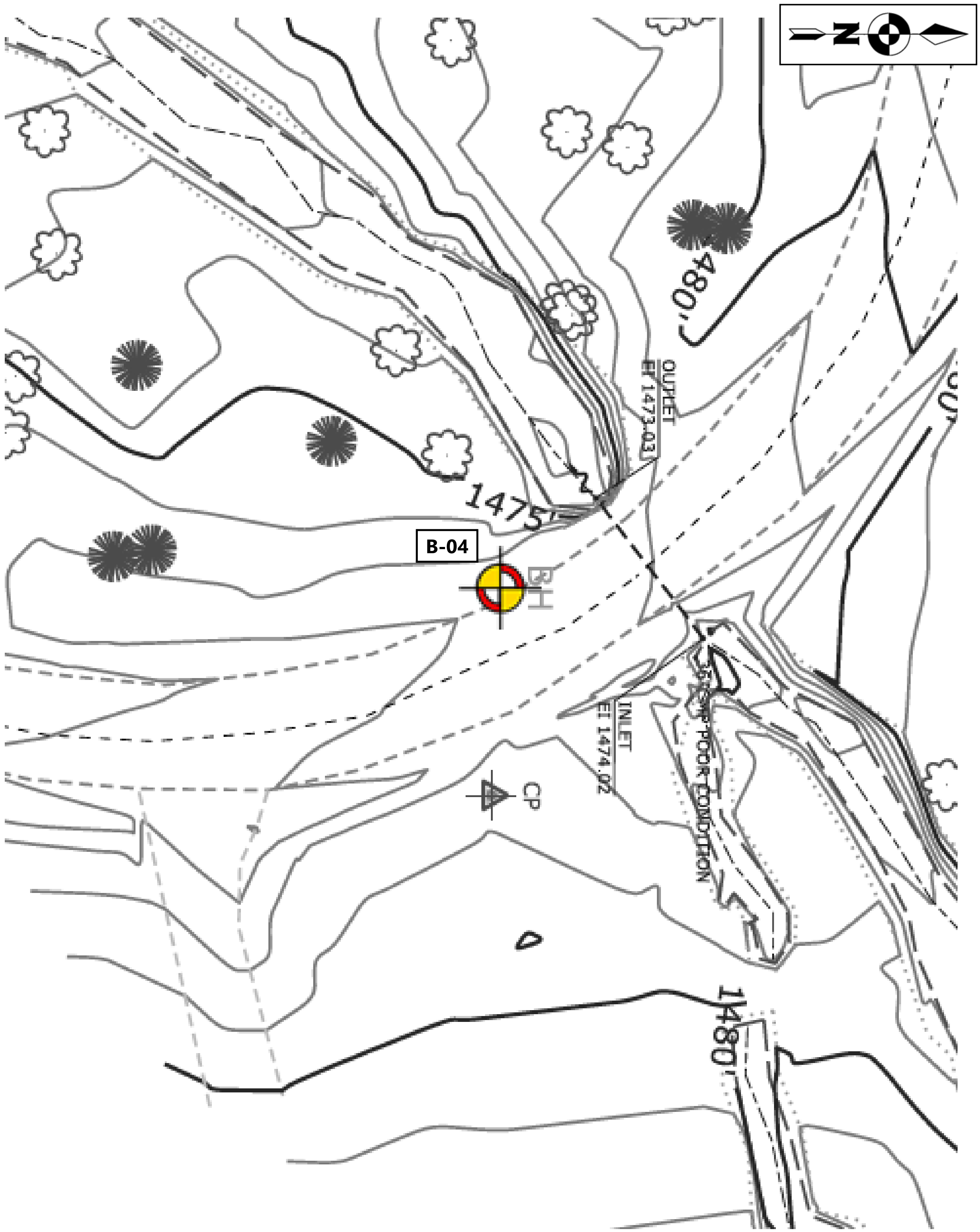
|---|---|---|---|
| S&ME | **BORING LOCATION PLAN – MASTER PIN 92/MILE POST 7.84** | SCALE: NOT TO SCALE | FIGURE NO. |
| | | DATE: 5/29/2024 | **5** |
| | FSR 221 PEAVINE SHEEDS CREEK<br>CONASAUGA, TENNESSEE | PROJECT NUMBER 23810276 | |




**NOTES:**
**IMAGERY PROVIDED BY WSP.**
**DRAWING FOR ILLUSTRATION PURPOSES ONLY**

**LEGEND:** **APPROXIMATE BORING LOCATION**

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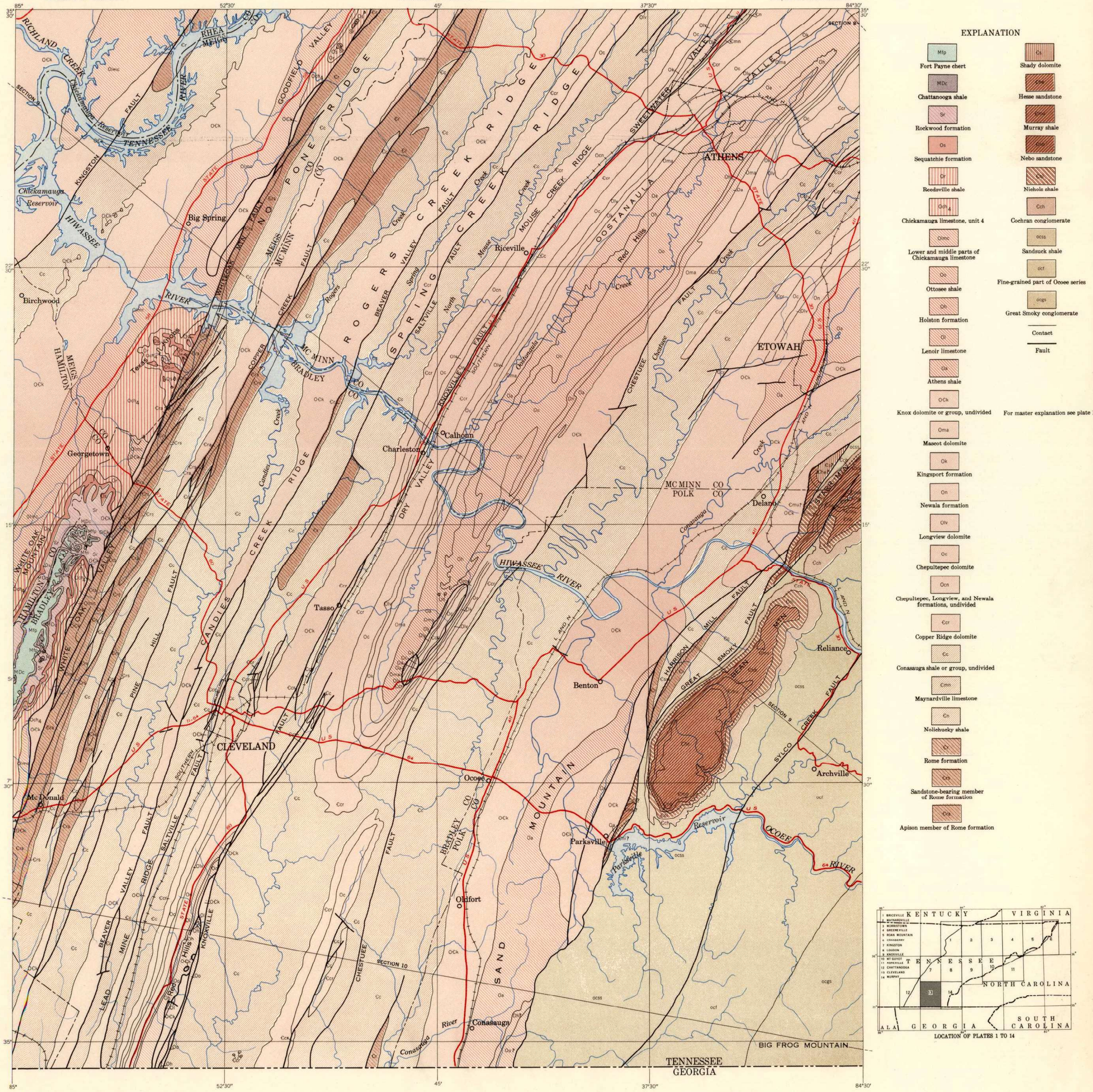
| **BORING LOCATION PLAN – MASTER PIN 179/MILE POST 19.302** | SCALE: NOT TO SCALE | DATE: 5/29/2024 | FIGURE NO. |
| FSR 221 PEAVINE SHEEDS CREEK<br>CONASAUGA, TENNESSEE | PROJECT NUMBER 23810276 | | **6** |

STATE OF TENNESSEE
DEPARTMENT OF CONSERVATION
DIVISION OF GEOLOGY

PREPARED IN COOPERATION WITH
UNITED STATES DEPARTMENT OF INTERIOR
GEOLOGICAL SURVEY

BULLETIN 58
PLATE 13, CLEVELAND




Base compiled and materials assembled by C. H. Heins, G. D. DeBuchananne and R. M. Richardson of the U. S. Geological Survey

Maps prepared for printing by N. B. Johnson, C. E. James, J. B. Canter, and others of the U. S. Geological Survey

Geology compiled by John Rodgers, U. S. Geological Survey
(For Description of sources see text)

# GEOLOGIC MAP OF EAST TENNESSEE: CLEVELAND

Scale $\frac{1}{125,000}$

1 0 5 10 Miles

# Appendix II

Soil Log Legend

Rock Core Log Legend

Test Boring Logs

Subsurface Profile - MP 7.84

# SOIL LOG LEGEND

## SOIL PROPERTY SYMBOLS

**N** - Standard Penetration, bpf
**NMC** - Natural Moisture Content, %
**F** - Fines Content, %
**LL** - Liquid Limit, %
**PL** - Plastic Limit, %
**PI** - Plasticity Index, %
**PPV** - Pocket Penetrometer Value, tsf
**Qu** - Unconfined Compressive Strength
**$\gamma$d** - Dry Unit Weight, pcf

**The STANDARD PENETRATION TEST (SPT)** as defined by ASTM D1586 (or AASHTO T206) is a method to obtain a disturbed soil sample for examination and testing and to obtain relative density and consistency information. A standard 1.4-inch I.D./2-inch O.D. split-barrel sampler is driven three 6-inch increments with a 140 lb. hammer freely falling 30 inches. The hammer can either be of a trip, free-fall design, or actuated by a rope and cathead. The SPT N Value is determined by adding the number of blows from the 2nd and 3rd 6-inch increments. A normalized blowcount ($N_{60}$) may be determined by the following equation: $N_{60}$ = [ Rig Energy Ratio (%) / 60 ] * N.

**SHELBY TUBE (ST)** samples are obtained by hydraulically pushing a thin-walled tube (typically 3-inches in diameter) to obtain a relatively undisturbed sample for testing of fine-grained soils to determine engineering properties such as strength, compressibility, permeability, and density. Shelby tubes are sampled in general accordance with ASTM D1587 (AASHTO T207).

SPT

ST

**Descriptive Order of Soil Strata:** Geologic Disposition (i.e., Fill, Colluvium, Alluvium, etc.), ASTM Group Name (ASTM Group Symbol), quantified/qualified soil constituents, misc. constituents, consistency/density, color, organic description, moisture. ASTM group classifications is determined per ASTM D2487 where lab testing has been performed and ASTM D2488 where lab testing has not been performed.

## ASTM GROUP NAME (SYMBOL) AND GRAPHIC

WELL GRADED GRAVEL (GW)
POORLY GRADED GRAVEL (GP)
WELL GRADED GRAVEL WITH SILT (GW-GM)
WELL GRADED GRAVEL WITH CLAY (GW-GC)
POORLY GRADED GRAVEL WITH SILT (GP-GM)
POORLY GRADED GRAVEL WITH CLAY (GP-GC)
SILTY GRAVEL (GM)
CLAYEY GRAVEL (GC)
CLAYEY GRAVEL WITH SILT (GC-GM)

WELL GRADED SAND (SW)
POORLY GRADED SAND (SP)
WELL GRADED SAND WITH SILT (SW-SM)
WELL GRADED SAND WITH CLAY (SW-SC)
POORLY GRADED SAND WITH SILT (SP-SM)
POORLY GRADED SAND WITH CLAY (SP-SC)
SILTY SAND (SM)
CLAYEY SAND (SC)
CLAYEY SAND WITH SILT (SC-SM)

LEAN CLAY (CL)
SILTY CLAY (CL-ML)
SILT (ML)
FAT CLAY (CH)
ELASTIC SILT (MH)
ORGANIC LOW PLASTICITY SILT OR CLAY (OL)
ORGANIC HIGH PLASTICITY SILT OR CLAY (OH)
PEAT (PT)

TOPSOIL
ASPHALT
BASE - CEMENT MODIFIED
BASE - CEMENT STABILIZED AGGREGATE
BASE - GRAVEL
CONCRETE
VOID / NO RECOVERY
IGM / PWR

## FINE-GRAINED SOIL (Relative Consistency)

| | N | PPV |
|---|---|---|
| Very Soft | < 2 bpf | < 0.25 tsf |
| Soft | 2 - 4 bpf | > 0.25 - 0.5 tsf |
| Firm | 5 - 8 bpf | > 0.5 - 1.0 tsf |
| Stiff | 9 - 15 bpf | > 1.0 - 2.0 tsf |
| Very Stiff | 16 - 30 bpf | > 2.0 - 4.0 tsf |
| Hard | > 30 bpf | > 4.0 tsf |

## COARSE-GRAINED SOIL (Relative Density)

| | N |
|---|---|
| Very Loose | < 5 bpf |
| Loose | 5 - 10 bpf |
| Medium Dense | 11 - 30 bpf |
| Dense | 31 - 50 bpf |
| Very Dense | > 50 bpf |

## MINOR CONSTITUENTS (% By Weight)

| | Percentage |
|---|---|
| Trace | 0% - 10% |
| Little | 11% - 20% |
| Some | 21% - 35% |
| “And” | $\geq$ 36% |

## ORGANIC CONTENT OF SOIL (Determined by ASTM D2974 or AASHTO T267)

| Classification | Percentage |
|---|---|
| With Organic Matter | 4% - 15% |
| Organic Soil | 16% - 30% |
| Peat | > 30% |

## MOISTURE CONDITION

| | |
|---|---|
| Dry | Absense of moisture, dusty, dry to touch |
| Moist | Damp but no visible water |
| Wet | Visible free water, usually soil is below water table |

**At Time of Drilling (ATD)** — Groundwater observation made anytime during the drilling process. Depending on time of reading and drilling methodologies, this value may be influenced by the drilling process.

**End of Drilling** — Groundwater measurement soon after the drilling processes are complete, and the borehole is at final depth. Drilling fluids, if introduced during drilling, may influence this measurement.

**After Drilling** — Groundwater measurements made in a borehole hours to days after drilling is complete. Depending on subsurface conditions, elapsed time, drilling process, etc. this observation may reflect a stabilized level.

REFERENCES:

FHWA NHI-16-072, Geotechnical Engineering Circular No. 5 “Geotechnical Site Characterization”
ASTM Specifications D2487 and D2488
DOT Specifications & Design Manuals from NC, SC, OH, MI, IN, PA, VA.

# ROCK CORE LOG LEGEND

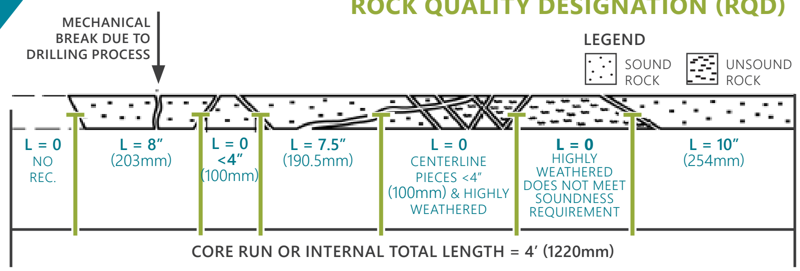
## ROCK CORE RECOVERY

| Rock Core Sample | Core Diameter (I.D.) | Inches |
|---|---|---|
| | BQ | 1-7/16 |
| | NQ | 1-7/8 |
| | HQ | 2-1/2 |

$$\text{REC} = \frac{\text{Length of Rock Core Recovered}}{\text{Length of Core Run}} \times 100$$

(Recovery)

## ROCK QUALITY DESIGNATION (RQD)




$$\text{RQD} = \left(\frac{\sum \text{Sound Core with Length (L)} \geq 4'' \text{ (100mm)}}{\text{Core Run or Interval Total Length}}\right) \times 100$$

(Equation)

$$\text{RQD} = \left(\frac{8'' + 7.5'' + 10''}{48''}\right) \times 100 = 53\% \text{ (FAIR)}$$

(Example)

### ROCK QUALITY

| Percent | Quality |
|---|---|
| >90–100 | Excellent |
| >75–90 | Good |
| >50–75 | Fair |
| >25–50 | Poor |
| 0–25 | Very Poor |

(ASTM D6032)

## GRAIN SIZE

| | |
|---|---|
| Very Fine-Grained | <0.003 in. (<0.075 mm) |
| Fine-Grained | 0.003 – 0.02 in. (0.075 – 0.425 mm) |
| Medium-Grained | 0.02 – 0.8 in. (0.425 – 2 mm) |
| Coarse-Grained | 0.8 – 2 in. (2 – 4.75 mm) |
| Very Coarse-Grained | >2 in. (>4.75 mm) |

## BEDDING

| | |
|---|---|
| Very Thickly Bedded | >3 ft. |
| Thickly Bedded | 3 ft. – 18 in. |
| Thinly Bedded | 18 in. – 2 in. |
| Very Thinly Bedded | 2 in. – 0.4 in. |
| Laminated | 0.4 in. – 0.1 in. |
| Thinly Laminated | <0.1 in. |

## FRACTURE RATE / SPACING

| | |
|---|---|
| Unfractured | >10 ft. |
| Intact | 10 ft. – 3 ft. |
| Slightly Fractured | 3 ft. – 1 ft. |
| Moderately Fractured | 12 in. – 4 in. |
| Fractured | 4 in. – 2 in. |
| Highly Fractured | < 2 in. |

## SURFACE ROUGHNESS

| | |
|---|---|
| Very Rough | Near vertical steps and ridges occur on the discontinuity surface. |
| Slightly Rough | Asperities on the discontinuity surface are distinguishable and can be felt. |
| Smooth | Surface appears smooth and feels so to the touch. |
| Slickensided | Surface has a smooth, glassy finish with visual evidence of striation. |

## WEATHERING

| | |
|---|---|
| Fresh | No visible sign of rock material weathering; slight discoloration on major discontinuity surfaces is possible. |
| Slightly Weathered | Discoloration indicates weathering of rock material and discontinuity surfaces. All rock material may be discolored by weathering and the external surface may be somewhat weaker than in its fresh condition. |
| Moderately Weathered | Less than half of the rock material is decomposed and/or disintegrated to a soil. Fresh or discolored rock is present either as a discontinuous framework or as corestones. A minimum 2 in. diameter sample cannot be broken readily by hand. |
| Highly Weathered | More than half the rock is decomposed or disintegrated to soil. Fresh or discolored rock is present either as a discontinuous framework or as corestones. A minimum 2 in. diameter sample can be broken readily by hand across the rock fabric. |
| Completely Weathered | All rock material is decomposed and/or disintegrated to soil. The original mass structure is largely still intact. Material can by granulated by hand. |
| Residual Soil | All rock material is converted to soil. Material can be easily broken apart by hand. |

## STRENGTH

| | | APPROX. UNCONFINED COMPRESSIVE STRENGTH (PSI) |
|---|---|---|
| Extremely Strong Rock | Specimen can only be chipped with firm blows from the hammer end of a geological hammer. | > 36,250 |
| Very Strong Rock | Specimen requires many firm blows from the hammer end of a geological hammer to fracture. | 36,250 – 14,500 |
| Strong Rock | Specimen requires more than one firm blow of the point of a geological hammer to fracture. | 14,500 – 7,250 |
| Medium Strong Rock | Specimen cannot be scraped or cut with a pocket knife. Specimen can be fractured with a single firm blow with a geological hammer point. | 7,250 – 3,500 |
| Weak Rock | Shallow cuts or scrapes can be made in a specimen with a pocket knife. A firm blow with a geological hammer creates shallow indents. | 3,500 – 725 |
| Very Weak Rock | Specimen crumbles under sharp blow with point of geological hammer and can be peeled with a pocket knife. | 725 – 150 |
| Extremely Weak Rock | Specimen can be indented by thumbnail. | 150 – 35 |

## HARDNESS

| | |
|---|---|
| Very Hard | Cannot be scratched with a pocket knife; leaves knife steel marks on surface. |
| Hard | Can be scratched by a pocket knife with difficulty; scratch produces little powder and is only faintly visible; trace of knife's steel may be visible. |
| Moderately Hard | Can be readily scratched by a pocket knife; scratch leaves a heavy trace of dust and scratch is readily visible after the powder has been blown away. |
| Low Hardness | Can be gouged deeply or carved with a pocket knife. |
| Friable | Easily crumbled by hand, pulverized or reduced to powder sand is too soft to be cut by a pocket knife. |
| Soft | Very weak plastic material. |

REFERENCES:

FHWA NHI-16-072, GEOTECHNICAL ENGINEERING CIRCULAR NO. 5 "GEOTECHNICAL SITE CHARACTERIZATION"
DOT SPECIFICATIONS & DESIGN MANUALS FROM NC, SC, OH, MI, IN, PA.

| PROJECT: | FSR 221 Peavine Sheeds Creek Road<br>Polk County, Tennessee<br>**S&ME Project No. 23810276** | **BORING LOG: B-01**<br>*Sheet 1 of 1* |
|---|---|---|

| | | |
|---|---|---|
| **DATE DRILLED:** 02/20/2024 | **ELEVATION:** 1135 ft | **NOTES:** FSR221 Mile 6.229, Master Pin 77 |
| **DRILL RIG:** CME 550 | **DATUM:** NAVD88 | |
| **DRILLER:** Anthony Hooks | **BORING DEPTH:** 10.3 ft | |
| **HAMMER TYPE:** Automatic hammer | **CLOSURE:** Cement Grout | |
| **DRILLING METHOD:** HQ CA, NQ | **LOGGED BY:** Pen Sibley | **LATITUDE:** 35.005756 **LONGITUDE:** -84.663646 |
| **SAMPLING METHOD:** CORE, SS | **PROJECT COORDINATE SYSTEM -** NAD 1983 StatePlane Tennessee FIPS 4100 Feet | |

| DEPTH (feet) | NOTES | Origin/Identifier | GRAPHIC | SAMPLE NO. (RECOVERY) | MATERIAL DESCRIPTION | BLOW COUNT DATA (SPT N-value) | STANDARD PENETRATION TEST DATA (△ % Fines, ○ NMC, ├┤ PL---LL; 20 40 60 80 100) | ELEVATION |
|---|---|---|---|---|---|---|---|---|
| 0 | Auger refusal at 0.3 feet | Residuum | | SS-1 | | 50/2"<br>N = 50/2" | | 1135 |
| | | 0.1 | | | TOPSOIL, 1 inch | | | |
| | | 0.3 | | | SANDY LEAN CLAY (CL), little rock fragments, hard, brown, moist | | | |
| | | | | Run-1<br>REC-97%<br>RQD-92% | SHALE, gray, continuous, excellent quality, medium-grained, fresh, very hard | | | |
| 5 | | | | | | | | 1130 |
| | | | | Run-2<br>REC-100%<br>RQD-90% | | | | |
| 10 | | 10.3 | | | Borehole terminated at 10.3 feet | | | 1125 |
| 15 | | | | | | | | 1120 |

| GROUNDWATER | | DATE | DEPTH (FT) | REMARKS |
|---|---|---|---|---|
| ATD | ▽ | 02/20/2024 | | Not encountered prior to coring |
| END OF DRILLING | ▼ | | | |
| AFTER DRILLING | ▼ | | | |
| AFTER DRILLING | ▼ | | | |

**GROUNDWATER DEPTHS ARE NOT EXACT AND MAY VARY SUBSTANTIALLY FROM THOSE INDICATED. ATD = AT TIME OF DRILLING**
**LL=Liquid Limit, PL = Plastic Limit, NMC = Natural Moisture Content, PPV = Pocket Penetrometer (tsf), PTV = Pocket Torvane (tsf),**
**AR = Auger Refusal, IGM = Intermediate Geomaterial**

| | | | |
|---|---|---|---|
| **Project:** | **FSR 221 Peavine Sheeds Creek Road**<br>**Polk County, Tennessee**<br>23810276 | **Core Photographs:** **B-01**<br>Sheet 1 of 1 | |
| **Date:** 02/20/2024 | **Elevation:** 1135 ft | Notes: | |
| **Photographed by:** Pen Sibley | Latitude: 35.005756 Longitude: -84.663646 | | |

B-01 Depth: 0.3 - 5.3 ft

B-01 Depth: 5.3 - 10.3 ft

**PROJECT:** FSR 221 Peavine Sheeds Creek Road
Polk County, Tennessee
**S&ME Project No. 23810276**

# BORING LOG: B-02

*Sheet 1 of 2*

| | |
|---|---|
| **DATE DRILLED:** 02/20/2024 | **ELEVATION:** 1049 ft |
| **DRILL RIG:** CME 550 | **DATUM:** NAVD88 |
| **DRILLER:** Anthony Hooks | **BORING DEPTH:** 20.2 ft |
| **HAMMER TYPE:** Automatic hammer | **CLOSURE:** Cement Grout |
| **DRILLING METHOD:** HQ CA, NQ | **LOGGED BY:** Pen Sibley |
| **SAMPLING METHOD:** SS, CORE | **PROJECT COORDINATE SYSTEM -** NAD 1983 StatePlane Tennessee FIPS 4100 Feet |

**NOTES:** FSR221, Mile 7.84, Master Pin 92 (North of failure)

**LATITUDE:** 34.999566 **LONGITUDE:** -84.646774

| DEPTH (feet) | NOTES | Origin/Identifier | SAMPLE NO. (RECOVERY) | MATERIAL DESCRIPTION | BLOW COUNT DATA (SPT N-value) | ELEVATION |
|---|---|---|---|---|---|---|
| 0 | 0.1 | Residuum | SS-1 | TOPSOIL, 1 inch | 31-50/2" N = 50/2" | 1049 |
| | | | | SILTY SAND WITH GRAVEL (SM), very dense, orange brown, moist | | |
| | 2.0 | | SS-2 | POORLY GRADED GRAVEL WITH CLAY AND SAND (GP-GC), some sand, very loose to dense, orange brown, wet | 19-2-2-5 N = 4 | |
| 5 | | | SS-3 | | 7-19-17-7 N = 36 | 1044 |
| | Poor recovery S-4; 6.0 | | SS-4 | CLAYEY GRAVEL (GC), some sand, medium dense to very dense, orange brown, wet | 45-7-7-7 N = 14 | |
| | Auger refusal at 9.0 feet | | SS-5 | | 7-50/3" N = 50/3" | |
| | 9.0 | | Run-1 REC-90% RQD-83% | SANDSTONE, gray, continuous, good quality to excellent quality, medium-grained, slightly weathered to fresh, very hard | | |
| 10 | | | | | | 1039 |
| | | | Run-2 REC-100% RQD-100% | | | |
| 15 | | | | | | 1034 |

STANDARD PENETRATION TEST DATA

△ % Fines
○ NMC
├┤ PL---LL

20 40 60 80 100

| GROUNDWATER | | DATE | DEPTH (FT) | REMARKS |
|---|---|---|---|---|
| ATD | ▽ | 02/20/2024 | | Not encountered prior to coring |
| END OF DRILLING | ▼ | | | |
| AFTER DRILLING | ▼ | | | |
| AFTER DRILLING | ▼ | | | |

**GROUNDWATER DEPTHS ARE NOT EXACT AND MAY VARY SUBSTANTIALLY FROM THOSE INDICATED. ATD = AT TIME OF DRILLING**
**LL=Liquid Limit, PL = Plastic Limit, NMC = Natural Moisture Content, PPV = Pocket Penetrometer (tsf), PTV = Pocket Torvane (tsf),**
**AR = Auger Refusal, IGM = Intermediate Geomaterial**

| | | | |
|---|---|---|---|
| **PROJECT:** | FSR 221 Peavine Sheeds Creek Road<br>Polk County, Tennessee<br>**S&ME Project No. 23810276** | | |
| **DATE DRILLED:** 02/20/2024 | **ELEVATION:** 1049 ft | **NOTES:** FSR221, Mile 7.84, Master Pin 92 (North of failure) | |
| **DRILL RIG:** CME 550 | **DATUM:** NAVD88 | | |
| **DRILLER:** Anthony Hooks | **BORING DEPTH:** 20.2 ft | | |
| **HAMMER TYPE:** Automatic hammer | **CLOSURE:** Cement Grout | | |
| **DRILLING METHOD:** HQ CA, NQ | **LOGGED BY:** Pen Sibley | **LATITUDE:** 34.999566 | **LONGITUDE:** -84.646774 |
| **SAMPLING METHOD:** CORE, SS | **PROJECT COORDINATE SYSTEM -** NAD 1983 StatePlane Tennessee FIPS 4100 Feet | | |

| DEPTH (feet) | NOTES | GRAPHIC | SAMPLE NO. (RECOVERY) | MATERIAL DESCRIPTION | ELEVATION |
|---|---|---|---|---|---|
| | | | Run-3<br>REC-100%<br>RQD-100% | SANDSTONE, gray, continuous, good quality to excellent quality, medium-grained, slightly weathered to fresh, very hard | |
| 20 | 20.2 | | | Borehole terminated at 20.2 feet | 1029 |
| 25 | | | | | 1024 |
| 30 | | | | | 1019 |

| GROUND WATER | | DATE/TIME | DEPTH (FT) | REMARKS |
|---|---|---|---|---|
| ATD | ▽ | 02/20/2024 | | Not encountered prior to coring |
| END OF DRILLING | ▼ | | | |
| AFTER DRILLING | ▼ | | | |
| AFTER DRILLING | ▼ | | | |

**GROUNDWATER DEPTHS ARE NOT EXACT AND MAY VARY SUBSTANTIALLY FROM THOSE INDICATED. ATD = AT TIME OF DRILLING**

| | | | |
|---|---|---|---|
| **Project:** | **FSR 221 Peavine Sheeds Creek Road**<br>**Polk County, Tennessee**<br>23810276 | **Core Photographs:** | **B-02** |
| | | Sheet 1 of 1 | |
| **Date:** 02/20/2024 | **Elevation:** 1049 ft | Notes: | |
| **Photographed by:** Pen Sibley | Latitude: 34.999566 Longitude: -84.646774 | | |

**B-02 Depth: 9.0 - 10.2 ft**

**B-02 Depth: 10.2 - 15.2 ft**

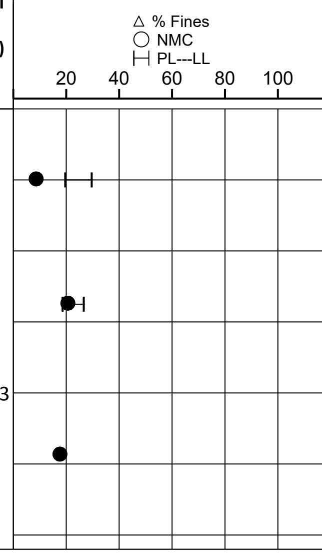
**B-02 Depth: 15.2 - 20.2 ft**

| PROJECT: | FSR 221 Peavine Sheeds Creek Road<br>Polk County, Tennessee<br>**S&ME Project No. 23810276** | **BORING LOG: B-03**<br>*Sheet 1 of 1* |
|---|---|---|

| | | |
|---|---|---|
| **DATE DRILLED:** 02/21/2024 | **ELEVATION:** 1044 ft | **NOTES:** FSR221, Mile 7.84, Master Pin 92 (South of failure) |
| **DRILL RIG:** CME 550 | **DATUM:** NAVD88 | |
| **DRILLER:** Anthony Hooks | **BORING DEPTH:** 15.5 ft | |
| **HAMMER TYPE:** Automatic hammer | **CLOSURE:** Cement Grout | |
| **DRILLING METHOD:** HQ CA, NQ | **LOGGED BY:** Pen Sibley | **LATITUDE:** 34.999450 **LONGITUDE:** -85.646658 |
| **SAMPLING METHOD:** SS, CORE | **PROJECT COORDINATE SYSTEM -** NAD 1983 StatePlane Tennessee FIPS 4100 Feet | |

| DEPTH (feet) | NOTES | Origin/Identifier | GRAPHIC | SAMPLE NO. (RECOVERY) | MATERIAL DESCRIPTION | BLOW COUNT DATA (SPT N-value) | ELEVATION |
|---|---|---|---|---|---|---|---|
| 0 – 0.2 | | Residuum | | | ASPHALT, 2 inches | | 1044 |
| 0.2 – 2.0 | | Residuum | | SS-1 | CLAYEY SAND WITH GRAVEL (SC), loose, orange brown, moist | 6-5-4-3<br>N = 9 | |
| 2.0 – 6.2 | Shelby tube attempted at 3.5 ft, no advancement due to gravel | Residuum | | SS-2 | CLAYEY GRAVEL WITH SAND (GC), medium dense, orange brown, moist | 11-11-10<br>N = 21 | |
| 5 | Auger refusal at 6.2 feet | | | SS-3 | | 48-11-7-50/3"<br>N = 18 | 1039 |
| 6.2 – 15.5 | | | | Run-1<br>REC-85%<br>RQD-77% | SANDSTONE, gray, fairly continuous to continuous, good quality to fair quality, medium-grained, slightly weathered to moderately weathered, very hard | | |
| 10 | | | | | | | 1034 |
| | | | | Run-2<br>REC-100%<br>RQD-60% | | | |
| 15 – 15.5 | | | | | Borehole terminated at 15.5 feet | | 1029 |

**STANDARD PENETRATION TEST DATA**




| GROUNDWATER | | DATE | DEPTH (FT) | REMARKS |
|---|---|---|---|---|
| ATD | | 02/21/2024 | | Not encountered prior to coring |
| END OF DRILLING | | | | |
| AFTER DRILLING | | | | |
| AFTER DRILLING | | | | |

**GROUNDWATER DEPTHS ARE NOT EXACT AND MAY VARY SUBSTANTIALLY FROM THOSE INDICATED. ATD = AT TIME OF DRILLING**
**LL=Liquid Limit, PL = Plastic Limit, NMC = Natural Moisture Content, PPV = Pocket Penetrometer (tsf), PTV = Pocket Torvane (tsf),**
**AR = Auger Refusal, IGM = Intermediate Geomaterial**

| | | | |
|---|---|---|---|
| **Project:** | **FSR 221 Peavine Sheeds Creek Road**<br>**Polk County, Tennessee**<br>23810276 | **Core Photographs:** | **B-03** |
| | | | Sheet 1 of 1 |
| **Date:** 02/21/2024 | **Elevation:** 1044 ft | Notes: | |
| **Photographed by:** Pen Sibley | Latitude: 34.999450 Longitude: -85.646658 | | |

**B-03 Depth: 6.2 - 10.5 ft**

**B-03 Depth: 10.5 - 15.5 ft**

| PROJECT: | FSR 221 Peavine Sheeds Creek Road<br>Polk County, Tennessee<br>**S&ME Project No. 23810276** | **BORING LOG: B-04**<br>*Sheet 1 of 2* |
|---|---|---|

| | | |
|---|---|---|
| **DATE DRILLED:** 02/22/2024 | **ELEVATION:** 1476 ft | **NOTES:** FSR221 Mile 19.302, Master Pin 179 |
| **DRILL RIG:** CME 550 | **DATUM:** NAVD88 | |
| **DRILLER:** Anthony Hooks | **BORING DEPTH:** 20.4 ft | |
| **HAMMER TYPE:** Automatic hammer | **CLOSURE:** Cement Grout | |
| **DRILLING METHOD:** HQ CA, NQ | **LOGGED BY:** Pen Sibley | **LATITUDE:** 35.052290 **LONGITUDE:** -84.532227 |
| **SAMPLING METHOD:** SS, CORE, UD | **PROJECT COORDINATE SYSTEM -** NAD 1983 StatePlane Tennessee FIPS 4100 Feet | |

| DEPTH (feet) | NOTES | Origin/Identifier | SAMPLE NO. (RECOVERY) | MATERIAL DESCRIPTION | BLOW COUNT DATA (SPT N-value) | ELEVATION |
|---|---|---|---|---|---|---|
| 0 | 0.1 | Residuum | SS-1 | TOPSOIL, 1 inch | 2-4-4-4<br>N = 8 | 1476 |
| | | | | SANDY SILT (ML), firm to stiff, gray with orange, moist | | |
| | Tube crushed | | UD-1<br>REC-84% | | | |
| 5 | | | SS-2 | | 4-2-7-22<br>N = 9 | 1471 |
| | 6.0 | | SS-3 | PWR, sampled as SANDY SILT (ML), hard, gray, relict structure, moist | 12-19-50/5"<br>N = 50/5" | |
| | | | SS-4 | | 8-24-50/5"<br>N = 50/5" | |
| 10 | Auger refusal at 10.1 feet | | | | | 1466 |
| | 10.1 | | Run-1<br>REC-86%<br>RQD-68% | SHALE, dark gray, fairly continuous to continuous, fair quality to excellent quality, moderately weathered to fresh, moderately hard | | |
| 15 | | | | | | 1461 |

STANDARD PENETRATION TEST DATA

△ % Fines
○ NMC
├┤ PL---LL

20 40 60 80 100

| GROUNDWATER | | DATE | DEPTH (FT) | REMARKS |
|---|---|---|---|---|
| ATD | | 02/22/2024 | | Not encountered prior to coring |
| END OF DRILLING | | | | |
| AFTER DRILLING | | | | |
| AFTER DRILLING | | | | |

**GROUNDWATER DEPTHS ARE NOT EXACT AND MAY VARY SUBSTANTIALLY FROM THOSE INDICATED. ATD = AT TIME OF DRILLING**
**LL=Liquid Limit, PL = Plastic Limit, NMC = Natural Moisture Content, PPV = Pocket Penetrometer (tsf), PTV = Pocket Torvane (tsf), AR = Auger Refusal, IGM = Intermediate Geomaterial**

| | | | |
|---|---|---|---|
| **PROJECT:** | FSR 221 Peavine Sheeds Creek Road<br>Polk County, Tennessee<br>**S&ME Project No. 23810276** | **BORING LOG: B-04**<br>*Sheet 2 of 2* | |
| **DATE DRILLED:** 02/22/2024 | **ELEVATION:** 1476 ft | **NOTES:** FSR221 Mile 19.302, Master Pin 179 | |
| **DRILL RIG:** CME 550 | **DATUM:** NAVD88 | | |
| **DRILLER:** Anthony Hooks | **BORING DEPTH:** 20.4 ft | | |
| **HAMMER TYPE:** Automatic hammer | **CLOSURE:** Cement Grout | | |
| **DRILLING METHOD:** HQ CA, NQ | **LOGGED BY:** Pen Sibley | **LATITUDE:** 35.052290 | **LONGITUDE:** -84.532227 |
| **SAMPLING METHOD:** CORE, SS, UD | **PROJECT COORDINATE SYSTEM -** NAD 1983 StatePlane Tennessee FIPS 4100 Feet | | |

| DEPTH (feet) | NOTES | GRAPHIC | SAMPLE NO. (RECOVERY) | MATERIAL DESCRIPTION | ELEVATION |
|---|---|---|---|---|---|
| | | | Run-2<br>REC-100%<br>RQD-93% | SHALE, dark gray, fairly continuous to continuous, fair quality to excellent quality, moderately weathered to fresh, moderately hard | |
| 20 | 20.4 | | | | 1456 |
| | | | | Borehole terminated at 20.4 feet | |
| 25 | | | | | 1451 |
| 30 | | | | | 1446 |

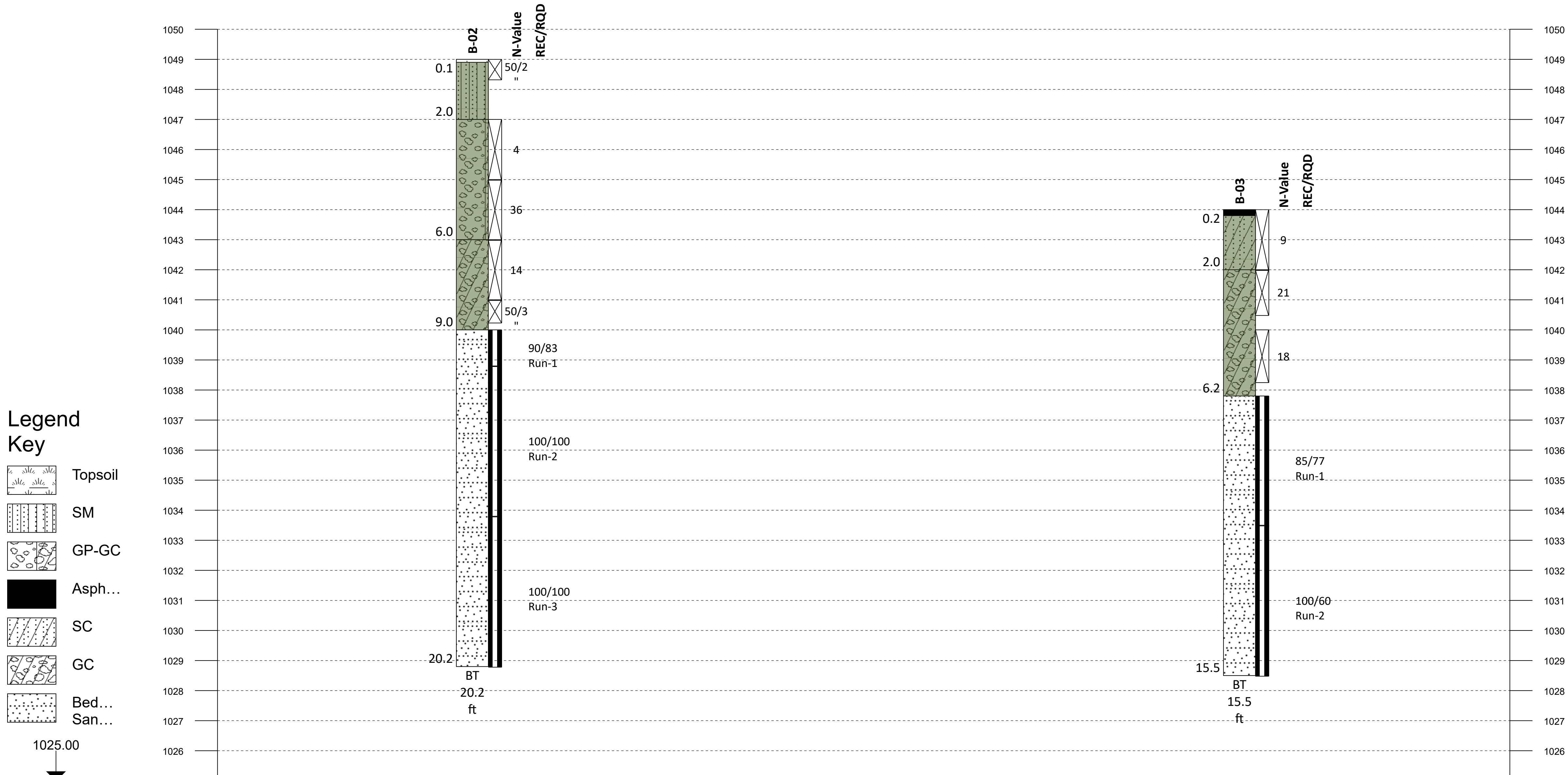
| GROUND WATER | | DATE/TIME | DEPTH (FT) | REMARKS |
|---|---|---|---|---|
| ATD | | 02/22/2024 | | Not encountered prior to coring |
| END OF DRILLING | | | | |
| AFTER DRILLING | | | | |
| AFTER DRILLING | | | | |

**GROUNDWATER DEPTHS ARE NOT EXACT AND MAY VARY SUBSTANTIALLY FROM THOSE INDICATED. ATD = AT TIME OF DRILLING**

| | | | |
|---|---|---|---|
| **Project:** | **FSR 221 Peavine Sheeds Creek Road**<br>**Polk County, Tennessee**<br>23810276 | **Core Photographs:** | **B-04** |
| | | Sheet 1 of 1 | |
| **Date:** 02/22/2024 | **Elevation:** 1476 ft | Notes: | |
| **Photographed by:** Pen Sibley | Latitude: 35.052290 Longitude: -84.532227 | | |

B-04 Depth: 10.1 - 15.4 ft

B-04 Depth: 15.4 - 20.4 ft




The depicted stratigraphy is shown for illustrative purposes only and is not warranted. Separations between different strata may be gradual and likely vary considerably from those shown. Profiles between nearby borings have been estimated using reasonable engineering care and judgement. The actual subsurface conditions will vary between boring locations.

AT TIME OF DRILLING
END OF DRILLING
AFTER DRILLING

Subsurface Profile - MP 7.84

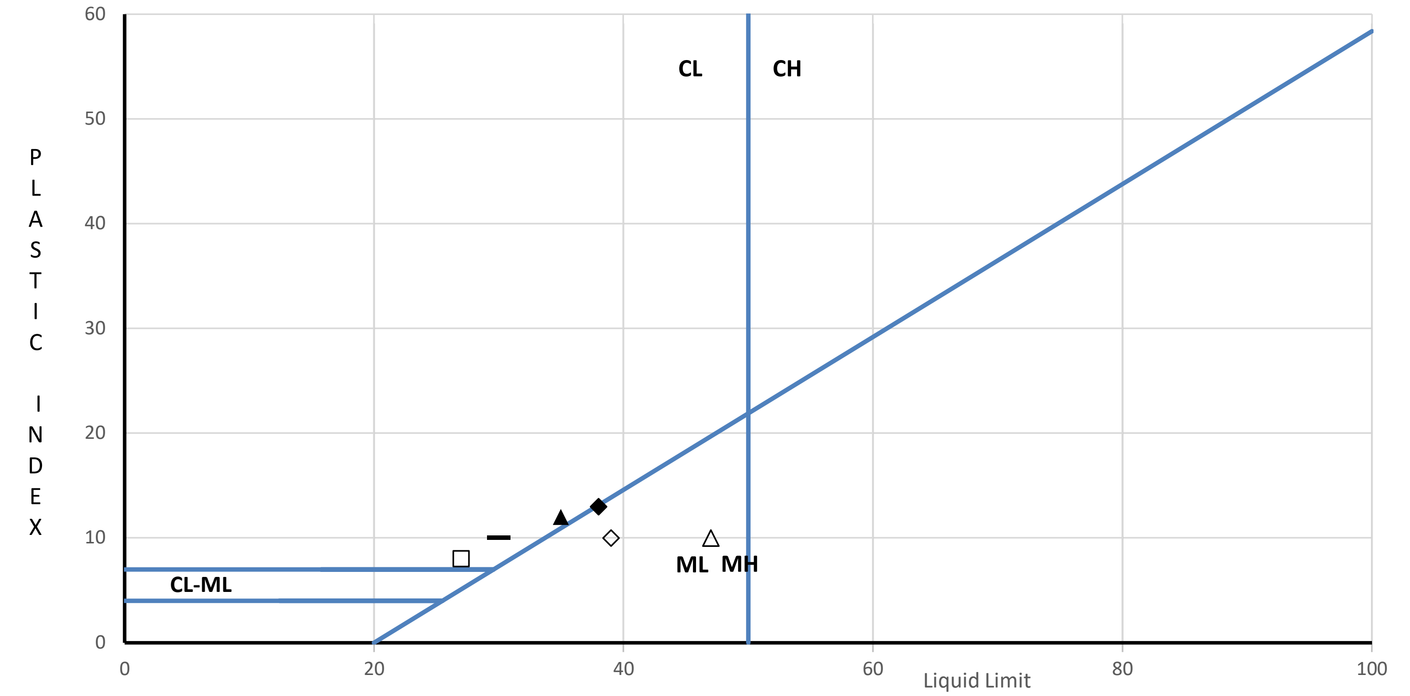
FSR 221 Peavine Sheeds Creek Road
Polk County, Tennessee

| | |
|---|---|
| SCALE: | 1:107 |
| DATE: | Aug 29, 2024 |
| PROJECT NUMBER: | 23810276 |

FIGURE NO
7

# Appendix III

Lab Test Results




| Specimen Identification | | | | MC | LL | PL | PI | Fines | Classification (symbol is based on minus 40 material only when no grain size information is present.) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | ID | No. | Top Depth | | | | | | Symbol | Name |
| | **B-01** | **SS-1** | **0** | **8.9** | | | | | | |
| ◆ | **B-02** | **SS-1** | **0** | | **38** | **25** | **13** | **43.2** | **SM** | **SILTY SAND WITH GRAVEL (A-6)** |
| ▲ | **B-02** | **SS-2** | **2** | | **35** | **23** | **12** | **9.4** | **GP-GC** | **POORLY GRADED GRAVEL WITH CLAY AND SAND (A-2-6)** |
| | **B-02** | **SS-3** | **4** | **5.2** | | | | | | |
| | **B-02** | **SS-4** | **6** | **1.6** | | | | | | |
| — | **B-03** | **SS-1** | **0** | | **30** | **20** | **10** | **29.4** | **SC** | **CLAYEY SAND WITH GRAVEL (A-2-4)** |
| □ | **B-03** | **SS-2** | **2** | | **27** | **19** | **8** | **15.3** | **GC** | **CLAYEY GRAVEL WITH SAND (A-2-4)** |
| ◇ | **B-04** | **SS-1** | **0** | **26.8** | **39** | **29** | **10** | **63.4** | **ML** | **SANDY SILT (A-4)** |
| △ | **B-04** | **SS-2** | **4** | **33.3** | **47** | **37** | **10** | **56.1** | **ML** | **SANDY SILT (A-5)** |
| | **B-04** | **SS-3** | **6** | **34.0** | | | | | | |

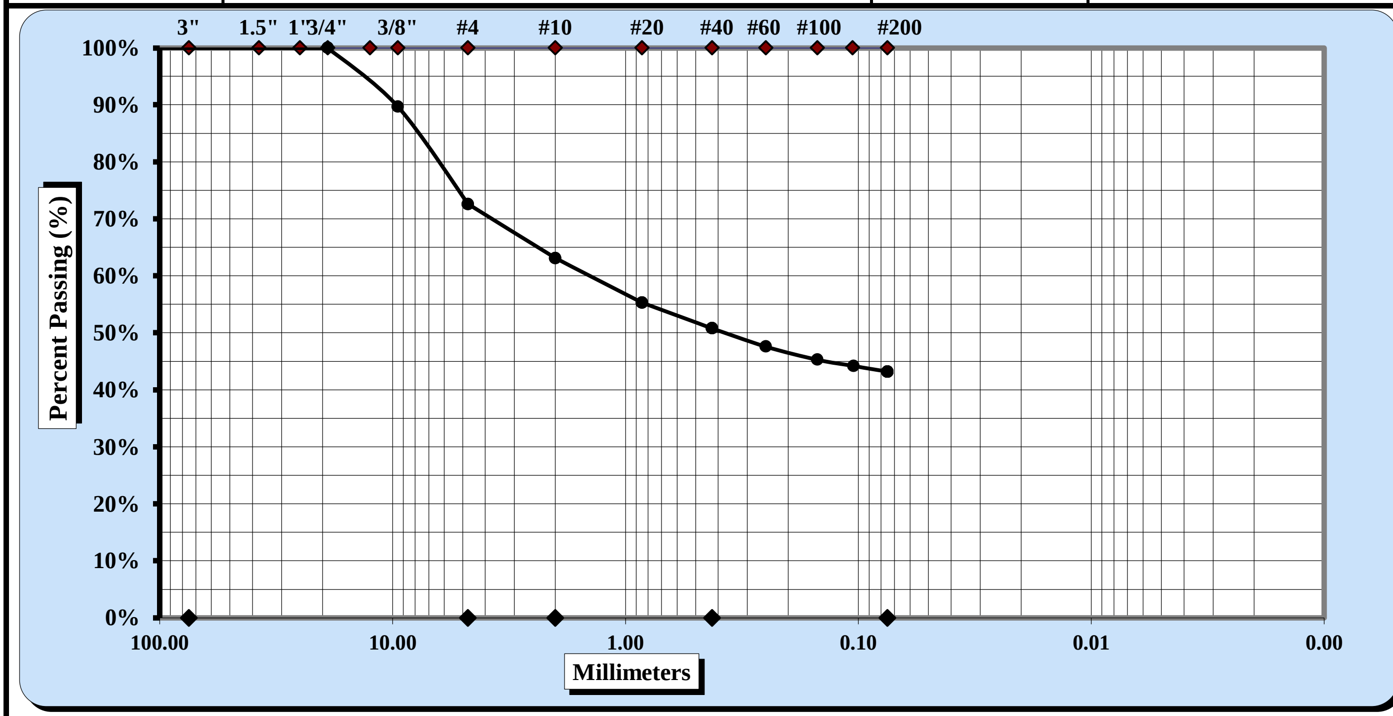
## INDEX TEST RESULTS

| | |
|---|---|
| Project Name | FR 221 Peavine Sheeds Creek |
| Project Number | 23810276 |
| Approved by | Date |
| | 3/29/2024 14:14 |

# AASHTO T88: Particle Size Analysis of Soils

| | |
|---|---|
| Report Number | CHAT_24001048 |
| Report Date | 3/29/2024 |
| Test Date | 3/25/2024 |
| Sample Date | 2/22/2024 |

| | | | |
|---|---|---|---|
| Project Number | 23810276 | | |
| Project Name | FR 221 Peavine Sheeds Creek | | |
| Client Name | WSP USA | | |
| Client Address | 13530 Dulles Technology Drive, Suite 300; Herndon, Virginia 20171 | | |
| KeyLAB ID | CHAT202403181 | Sample Type | SS |
| Location ID | B-02 | Sample Top Depth | 0 |
| Sample Reference | SS-1 | Sample Base Depth | 2 |
| Description | Yellowish Red - Brown | Method | AASHTO T 88 |
| Classification | SILTY SAND WITH GRAVEL | | |




ASTM PARTICLE SIZE DEFINITIONS

| | | | |
|---|---|---|---|
| Cobbles | < 300 mm (12") and > 75 mm (3") | Medium Sand | < 2.00 mm and > 0.425 mm (#40) |
| Gravel | < 75 mm and > 4.75 mm (#4) | Fine Sand | < 0.425 mm and > 0.075 mm |
| Coarse Sand | < 4.75 mm and >2.00 mm (#10) | Silt & Clay | < 0.075 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum Particle Size | **19 mm** | Coarse Sand | 9.5 | Fine Sand | 7.6 |
| Gravel | 27.4 | Medium Sand | 12.3 | Silt & Clay | 43.2 |
| Liquid Limit | 38 | Plastic Limit | 25 | Plastic Index | 13 |

Description of Sand & Gravel Particles: Rounded ☐ Angular ☒

Hard & Durable ☒ Soft ☐ Weathered & Friable ☐

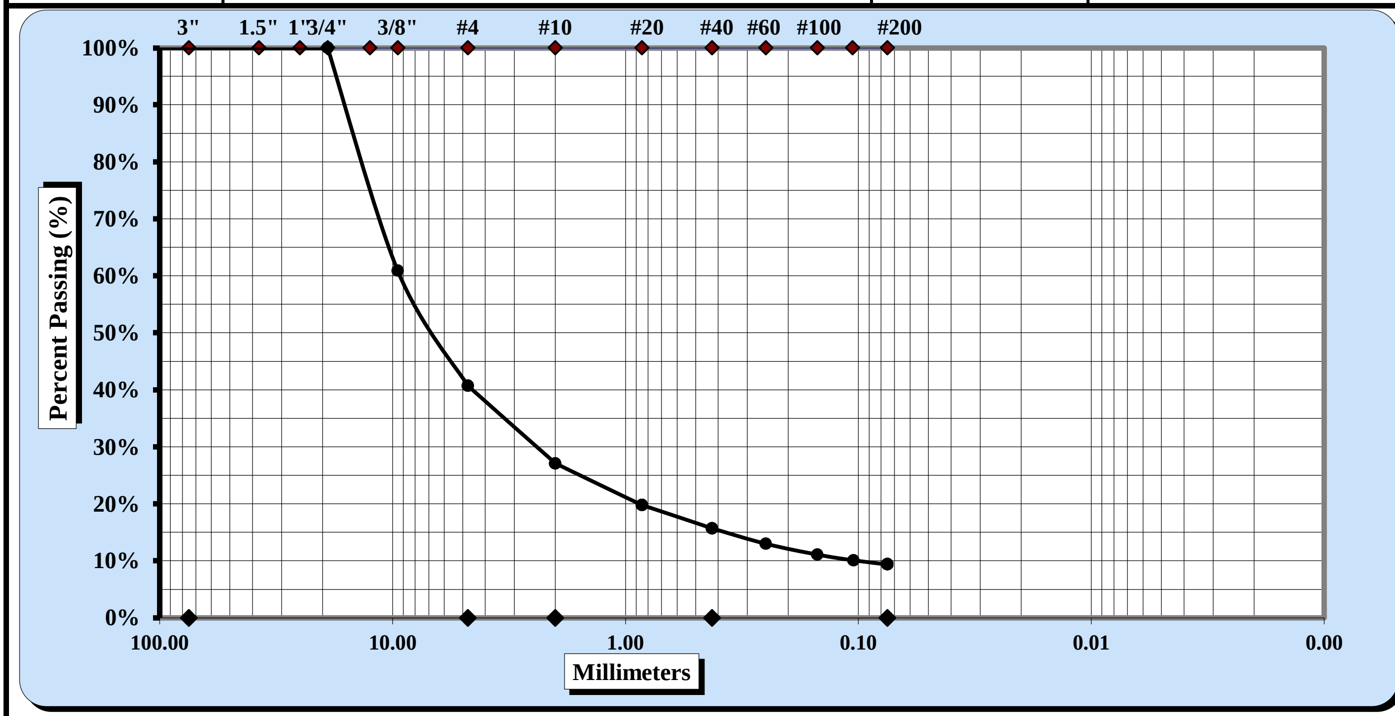
*References / Comments / Deviations:*



| | | |
|---|---|---|
| TThompson | TThompson | |
| *Tested by* | *Approved by* | *Signature* |
| *Chattanooga* | *4291 Highway 58, Suite 101, Chattanooga, TN 37416* | |

| | | | |
|---|---|---|---|
| | **AASHTO T88: Particle Size Analysis of Soils** | Report Number | **CHAT_24001049** |
| | | Report Date | **3/29/2024** |
| | | Test Date | **3/25/2024** |
| | | Sample Date | **2/22/2024** |

| | | | |
|---|---|---|---|
| Project Number | 23810276 | | |
| Project Name | FR 221 Peavine Sheeds Creek | | |
| Client Name | WSP USA | | |
| Client Address | 13530 Dulles Technology Drive, Suite 300; Herndon, Virginia 20171 | | |
| KeyLAB ID | CHAT202403182 | Sample Type | SS |
| Location ID | B-02 | Sample Top Depth | 2 |
| Sample Reference | SS-2 | Sample Base Depth | 4 |
| Description | Brown | Method | AASHTO T 88 |
| Classification | POORLY GRADED GRAVEL WITH CLAY AND SAND | | |




| ASTM PARTICLE SIZE DEFINITIONS | | | |
|---|---|---|---|
| Cobbles | < 300 mm (12") and > 75 mm (3") | Medium Sand | < 2.00 mm and > 0.425 mm (#40) |
| Gravel | < 75 mm and > 4.75 mm (#4) | Fine Sand | < 0.425 mm and > 0.075 mm |
| Coarse Sand | < 4.75 mm and >2.00 mm (#10) | Silt & Clay | < 0.075 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum Particle Size | **19 mm** | Coarse Sand | 13.6 | Fine Sand | 6.3 |
| Gravel | 59.3 | Medium Sand | 11.4 | Silt & Clay | 9.4 |
| Liquid Limit | 35 | Plastic Limit | 23 | Plastic Index | 12 |
| | **Cc =** | **6.28** | **Cu =** | **92.04** | |

Description of Sand & Gravel Particles: Rounded ☐ Angular ☒

Hard & Durable ☒ Soft ☐ Weathered & Friable ☐

*References / Comments / Deviations:*

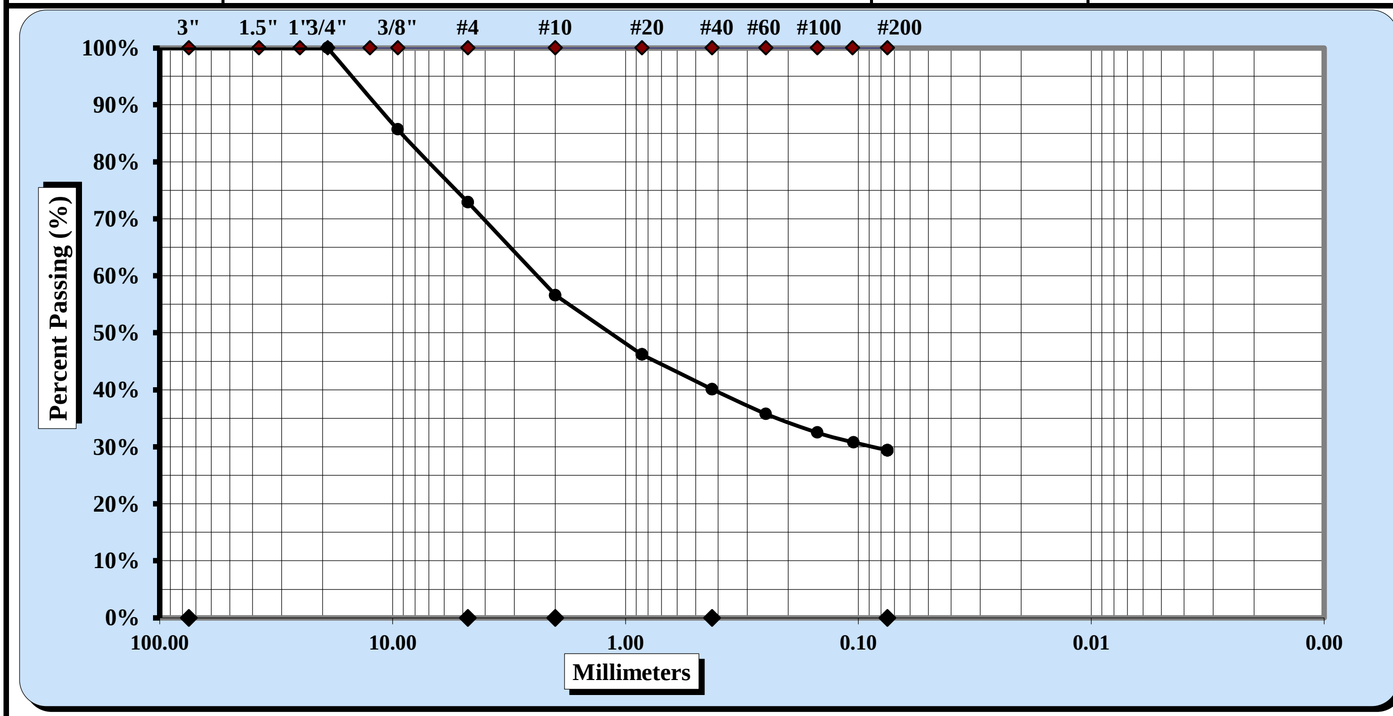


| | | |
|---|---|---|
| TThompson | TThompson | |
| *Tested by* | *Approved by* | *Signature* |
| *Chattanooga* | *4291 Highway 58, Suite 101, Chattanooga, TN 37416* | |

| | **AASHTO T88: Particle Size Analysis of Soils** | Report Number | **CHAT_24001050** |
|---|---|---|---|
| | | Report Date | **3/29/2024** |
| | | Test Date | **3/25/2024** |
| | | Sample Date | **2/22/2024** |

| | |
|---|---|
| Project Number | 23810276 |
| Project Name | FR 221 Peavine Sheeds Creek |
| Client Name | WSP USA |
| Client Address | 13530 Dulles Technology Drive, Suite 300; Herndon, Virginia 20171 |

| | | | |
|---|---|---|---|
| KeyLAB ID | CHAT202403185 | Sample Type | SS |
| Location ID | B-03 | Sample Top Depth | 0 |
| Sample Reference | SS-1 | Sample Base Depth | 2 |
| Description | Yellowish Red - Brown | Method | AASHTO T 88 |
| Classification | CLAYEY SAND WITH GRAVEL | | |




| ASTM PARTICLE SIZE DEFINITIONS | | | |
|---|---|---|---|
| Cobbles | < 300 mm (12") and > 75 mm (3") | Medium Sand | < 2.00 mm and > 0.425 mm (#40) |
| Gravel | < 75 mm and > 4.75 mm (#4) | Fine Sand | < 0.425 mm and > 0.075 mm |
| Coarse Sand | < 4.75 mm and >2.00 mm (#10) | Silt & Clay | < 0.075 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum Particle Size | **19 mm** | Coarse Sand | 16.3 | Fine Sand | 10.7 |
| Gravel | 27.1 | Medium Sand | 16.5 | Silt & Clay | 29.4 |
| Liquid Limit | 30 | Plastic Limit | 20 | Plastic Index | 10 |

Description of Sand & Gravel Particles: Rounded ☐ Angular ☒

Hard & Durable ☒ Soft ☐ Weathered & Friable ☐

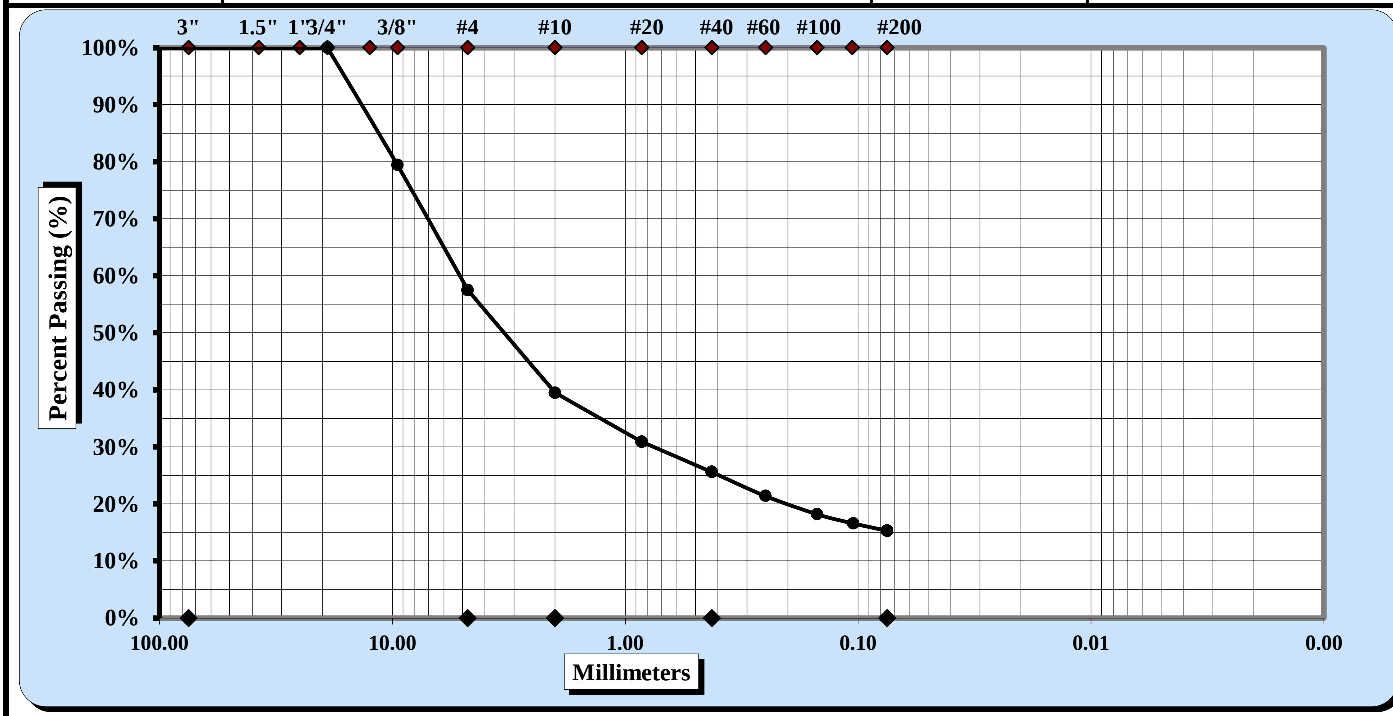
*References / Comments / Deviations:*



| | | |
|---|---|---|
| TThompson | TThompson | |
| *Tested by* | *Approved by* | *Signature* |
| *Chattanooga* | *4291 Highway 58, Suite 101, Chattanooga, TN 37416* | |

| | AASHTO T88: Particle Size Analysis of Soils | Report Number | CHAT_24001051 |
|---|---|---|---|
| | | Report Date | 3/29/2024 |
| | | Test Date | 3/25/2024 |
| | | Sample Date | 2/22/2024 |

| | | | |
|---|---|---|---|
| Project Number | 23810276 | | |
| Project Name | FR 221 Peavine Sheeds Creek | | |
| Client Name | WSP USA | | |
| Client Address | 13530 Dulles Technology Drive, Suite 300; Herndon, Virginia 20171 | | |
| KeyLAB ID | CHAT202403186 | Sample Type | SS |
| Location ID | B-03 | Sample Top Depth | 2 |
| Sample Reference | SS-2 | Sample Base Depth | 4 |
| Description | Yellowish Brown | Method | AASHTO T 88 |
| Classification | CLAYEY GRAVEL WITH SAND | | |




| ASTM PARTICLE SIZE DEFINITIONS | | | |
|---|---|---|---|
| Cobbles | < 300 mm (12") and > 75 mm (3") | Medium Sand | < 2.00 mm and > 0.425 mm (#40) |
| Gravel | < 75 mm and > 4.75 mm (#4) | Fine Sand | < 0.425 mm and > 0.075 mm |
| Coarse Sand | < 4.75 mm and >2.00 mm (#10) | Silt & Clay | < 0.075 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum Particle Size | **19 mm** | Coarse Sand | 18.0 | Fine Sand | 10.3 |
| Gravel | 42.5 | Medium Sand | 13.9 | Silt & Clay | 15.3 |
| Liquid Limit | 27 | Plastic Limit | 19 | Plastic Index | 8 |

Description of Sand & Gravel Particles: Rounded ☐ Angular ☒

Hard & Durable ☒ Soft ☐ Weathered & Friable ☐

*References / Comments / Deviations:*



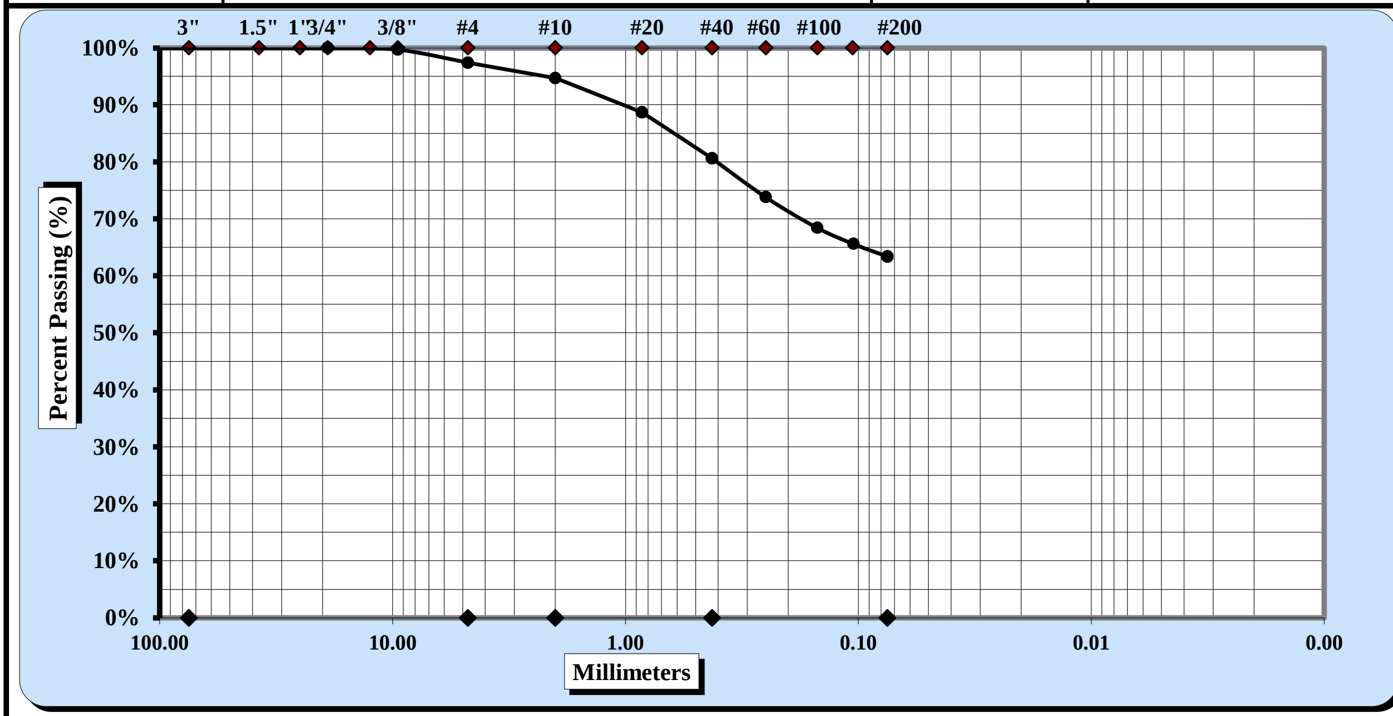
| TThompson | TThompson | |
|---|---|---|
| *Tested by* | *Approved by* | *Signature* |
| *Chattanooga* | *4291 Highway 58, Suite 101, Chattanooga, TN 37416* | |

# AASHTO T88: Particle Size Analysis of Soils

| | |
|---|---|
| Report Number | CHAT_24001052 |
| Report Date | 3/29/2024 |
| Test Date | 3/25/2024 |
| Sample Date | 2/22/2024 |

| | |
|---|---|
| Project Number | 23810276 |
| Project Name | FR 221 Peavine Sheeds Creek |
| Client Name | WSP USA |
| Client Address | 13530 Dulles Technology Drive, Suite 300; Herndon, Virginia 20171 |

| | | | |
|---|---|---|---|
| KeyLAB ID | CHAT202403187 | Sample Type | SS |
| Location ID | B-04 | Sample Top Depth | 0 |
| Sample Reference | SS-1 | Sample Base Depth | 2 |
| Description | Brown - Dark Brown | Method | AASHTO T 88 |
| Classification | SANDY SILT | | |




| ASTM PARTICLE SIZE DEFINITIONS | | | |
|---|---|---|---|
| Cobbles | < 300 mm (12") and > 75 mm (3") | Medium Sand | < 2.00 mm and > 0.425 mm (#40) |
| Gravel | < 75 mm and > 4.75 mm (#4) | Fine Sand | < 0.425 mm and > 0.075 mm |
| Coarse Sand | < 4.75 mm and >2.00 mm (#10) | Silt & Clay | < 0.075 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum Particle Size | **9.5 mm** | Coarse Sand | 2.7 | Fine Sand | 17.2 |
| Gravel | 2.6 | Medium Sand | 14.1 | Silt & Clay | 63.4 |
| Liquid Limit | 39 | Plastic Limit | 29 | Plastic Index | 10 |

Description of Sand & Gravel Particles: Rounded ☐ Angular ☒

Hard & Durable ☐ Soft ☐ Weathered & Friable ☒

*References / Comments / Deviations:*



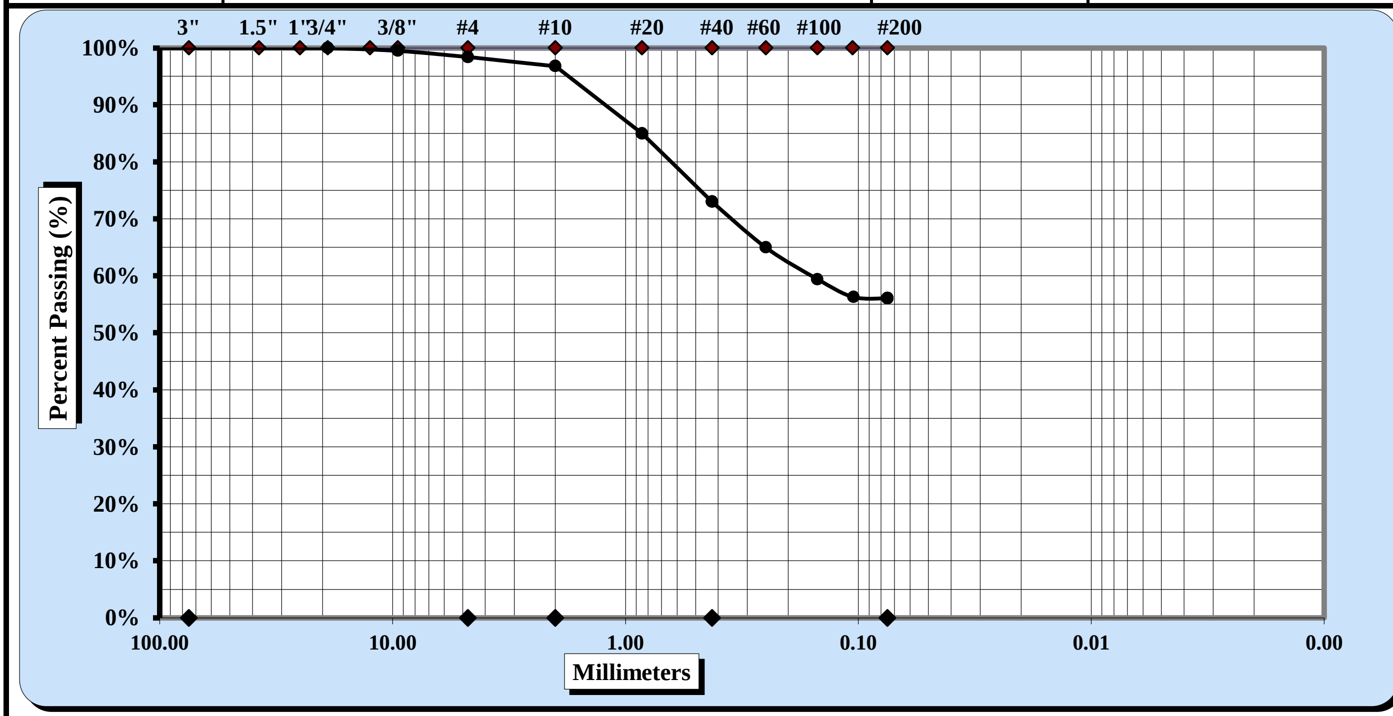
| | | |
|---|---|---|
| TThompson | TThompson | |
| *Tested by* | *Approved by* | *Signature* |
| *Chattanooga* | *4291 Highway 58, Suite 101, Chattanooga, TN 37416* | |

# AASHTO T88: Particle Size Analysis of Soils

| | |
|---|---|
| Report Number | CHAT_24001053 |
| Report Date | 3/29/2024 |
| Test Date | 3/25/2024 |
| Sample Date | 2/22/2024 |

| | |
|---|---|
| Project Number | 23810276 |
| Project Name | FR 221 Peavine Sheeds Creek |
| Client Name | WSP USA |
| Client Address | 13530 Dulles Technology Drive, Suite 300; Herndon, Virginia 20171 |

| | | | |
|---|---|---|---|
| KeyLAB ID | CHAT202403188 | Sample Type | SS |
| Location ID | B-04 | Sample Top Depth | 4 |
| Sample Reference | SS-2 | Sample Base Depth | 6 |
| Description | Black - Gray | Method | AASHTO T 88 |
| Classification | SANDY SILT | | |




ASTM PARTICLE SIZE DEFINITIONS

| | | | |
|---|---|---|---|
| Cobbles | < 300 mm (12") and > 75 mm (3") | Medium Sand | < 2.00 mm and > 0.425 mm (#40) |
| Gravel | < 75 mm and > 4.75 mm (#4) | Fine Sand | < 0.425 mm and > 0.075 mm |
| Coarse Sand | < 4.75 mm and >2.00 mm (#10) | Silt & Clay | < 0.075 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum Particle Size | **9.5 mm** | Coarse Sand | 1.6 | Fine Sand | 16.9 |
| Gravel | 1.6 | Medium Sand | 23.8 | Silt & Clay | 56.1 |
| Liquid Limit | 47 | Plastic Limit | 37 | Plastic Index | 10 |

Description of Sand & Gravel Particles: Rounded ☐ Angular ☒

Hard & Durable ☐ Soft ☒ Weathered & Friable ☐

*References / Comments / Deviations:*



| | | |
|---|---|---|
| TThompson | TThompson | |
| *Tested by* | *Approved by* | *Signature* |
| *Chattanooga* | *4291 Highway 58, Suite 101, Chattanooga, TN 37416* | |

# UNCONFINED COMPRESSION
# (ASTM D7012 Method C)

S&ME, Inc. - Knoxville 1413 Topside Road, Louisville, TN 37777

Project Name: FR 221 Peavine Sheeds Creek
Project Number: 23810276

Report Date: June 6, 2024
Reviewed By: Tyler Copeman

| Boring No. | Sample No. | Depth (ft) | Dimensions, in. | | Shape (See Key) | Area ($in^2$) | Unit Weight ($lbs/ft^3$) | Loading Rate (psi/sec) | Maximum Load (lbs) | Strength (psi) | Moisture (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Length | Diameter | | | | | | | |
| B-02 | Run-1 | 9.1 - 9.5 | 4.53 | 1.98 | A | 3.08 | 169.3 | 99 | 109,341 | 35,500 | 0.0 |
| B-03 | Run-1 | 9.7 - 10.1 | 4.65 | 1.98 | A | 3.08 | 169.1 | 91 | 63,086 | 20,482 | 0.1 |

NOTES: Effective (as received) unit weight as determined by RTH 109-93.
Loading rates were selected to target reaching failure between 2 and 15 minutes.
Test results for specimens not meeting the requirements of ASTM D4543-19 may differ from a test specimen that meets the requirements of ASTM D4543.

SHAPE KEY

ASTM D4543-19 Standard Practice for Preparing Rock Core as Cylindrical Test Specimens and Verifying Conformance to Dimensional and Shape Tolerance Section 1.2 - "Rock is a complex engineering material that can vary greatly as a function of lithology, stress history, weathering, moisture content and chemistry, and other natural geologic processes. As such, it is not always possible to obtain or prepare rock core specimens that satisfy the desirable tolerances given in this practice. Most commonly, this situation presents itself with weaker, more porous, and poorly cemented rock types and rock types containing significant or weak (or both) structural features. For rock types which are difficult to prepare, all reasonable efforts shall be made to prepare a specimen in accordance with this practice and for the intended test procedure. However, when it has been determined by trial and error that this is not possible, prepare the rock specimen to the closest tolerances practicable and consider this to be the best effort and report it as such and if allowable or necessary for the intended test, capping the ends of the specimen as discussed in this practice is permitted."

A Test specimen measurements met the desired shape tolerances of ASTM D4543-19 (side straightness, end flatness & parallelism, and end perpendicularity to axis)

B Test specimen measurements met the desired shape tolerances of ASTM D4543-19 for end flatness & parallelism, and end perpendicularity to axis. Specimen did not meet the desired tolerance for side straightness. Specimen prepared to closest tolerances practicable.

C Test specimen measurements met the desired shape tolerances of ASTM D4543-19 for end flatness & parallelism. Specimen did not meet the desired tolerances for side straightness and end perpendicularity to axis. Specimen prepared to closest tolerances practicable.

D Test specimen measurements met the desired shape tolerances of ASTM D4543-19 for end flatness. Specimen did not meet the desired tolerances for side straightness, parallelism and end perpendicularity to axis. Specimen prepared to closest tolerances practicable.

E Test specimen measurements met the desired shape tolerances of ASTM D4543-19 for end flatness and end perpendicularity to axis. Specimen did not meet the desired tolerance for side straightness and parallelism. Specimen prepared to closest tolerances practicable.

Date: 06/06/2024

Photographer: Ryan Skinner

| | | |
|---|---|---|
| **1** | **Location / Orientation** | B-02, Run-1 (9.1' – 9.5') |
| | **Remarks** | Unconfined Compressive Strength of Rock Core Specimen Before/After (ASTM D7012 Method C) |

Date: 06/06/2024

Photographer: Ryan Skinner

| | | |
|---|---|---|
| **2** | **Location / Orientation** | B-03, Run-1 (9.7' – 10.1') |
| | **Remarks** | Unconfined Compressive Strength of Rock Core Specimen Before/After (ASTM D7012 Method C) |

# Soil Analysis Lab Results

Client: S&ME, Inc.
Job Name: FSR 221 Peavine Sheeds Creek Road
Client Job Number: 23810276
Project X Job Number: S240529C
May 30, 2024

| | Method | ASTM D4327 | | ASTM D4327 | | ASTM G187 | | ASTM G51 |
|---|---|---|---|---|---|---|---|---|
| **Bore# / Description** | **Depth** | **Sulfates** $SO_4^{2-}$ | | **Chlorides** $Cl^-$ | | **Resistivity** As Rec'd \| Minimum | | **pH** |
| | (ft) | (mg/kg) | (wt%) | (mg/kg) | (wt%) | (Ω-cm) | (Ω-cm) | |
| B-01 SS-1 | 0-2 | 126.8 | 0.0127 | 24.5 | 0.0024 | NT | NT | 7.7 |
| B-02 SS-5 | 8-10 | 362.4 | 0.0362 | 135.8 | 0.0136 | >737,000 | 7,370 | 6.6 |
| B-03 SS-3 | 4-6 | 62.6 | 0.0063 | 21.4 | 0.0021 | >737,000 | 10,650 | 6.7 |
| B-04 SS-3 | 6-8 | 138.7 | 0.0139 | 32.7 | 0.0033 | 3,216 | 1,809 | 7.1 |
| B-04 SS-4 | 6-8 | 394.8 | 0.0395 | 51.3 | 0.0051 | 2,613 | 1,206 | 4.0 |

Cations and Anions, except Sulfide and Bicarbonate, tested with Ion Chromatography
mg/kg = milligrams per kilogram (parts per million) of dry soil weight
ND = 0 = Not Detected | NT = Not Tested | Unk = Unknown
Chemical Analysis performed on 1:3 Soil-To-Water extract
PPM = mg/kg (soil) = mg/L (Liquid)

**Note**: Sometimes a bad sulfate hit is a contaminated spot. Typical fertilizers are Potassium chloride, ammonium sulfate or ammonium sulfate nitrate (ASN). So this is another reason why testing full corrosion series is good because we then have the data to see if those other ingredients are present meaning the soil sample is just fertilizer-contaminated soil. This can happen often when the soil samples collected are simply surface scoops. This is why it's best to dig in a foot, throw away the top and test the deeper stuff. Dairy farms are also notorious for these items.

If one sample pops up much more corrosive than all others, we would recommend collecting more samples surrounding the problem sample location to determine if the peak is isolated to it. This allows us to conclude it was a contaminated sample and able to declare it an outlier.

Try out our new online forms: SOIL CORROSIVITY & THERMAL RESISTIVITY LAB REQUEST FORM & IN-SITU WENNER 4 PIN QUOTE REQUEST FORM

# Appendix IV

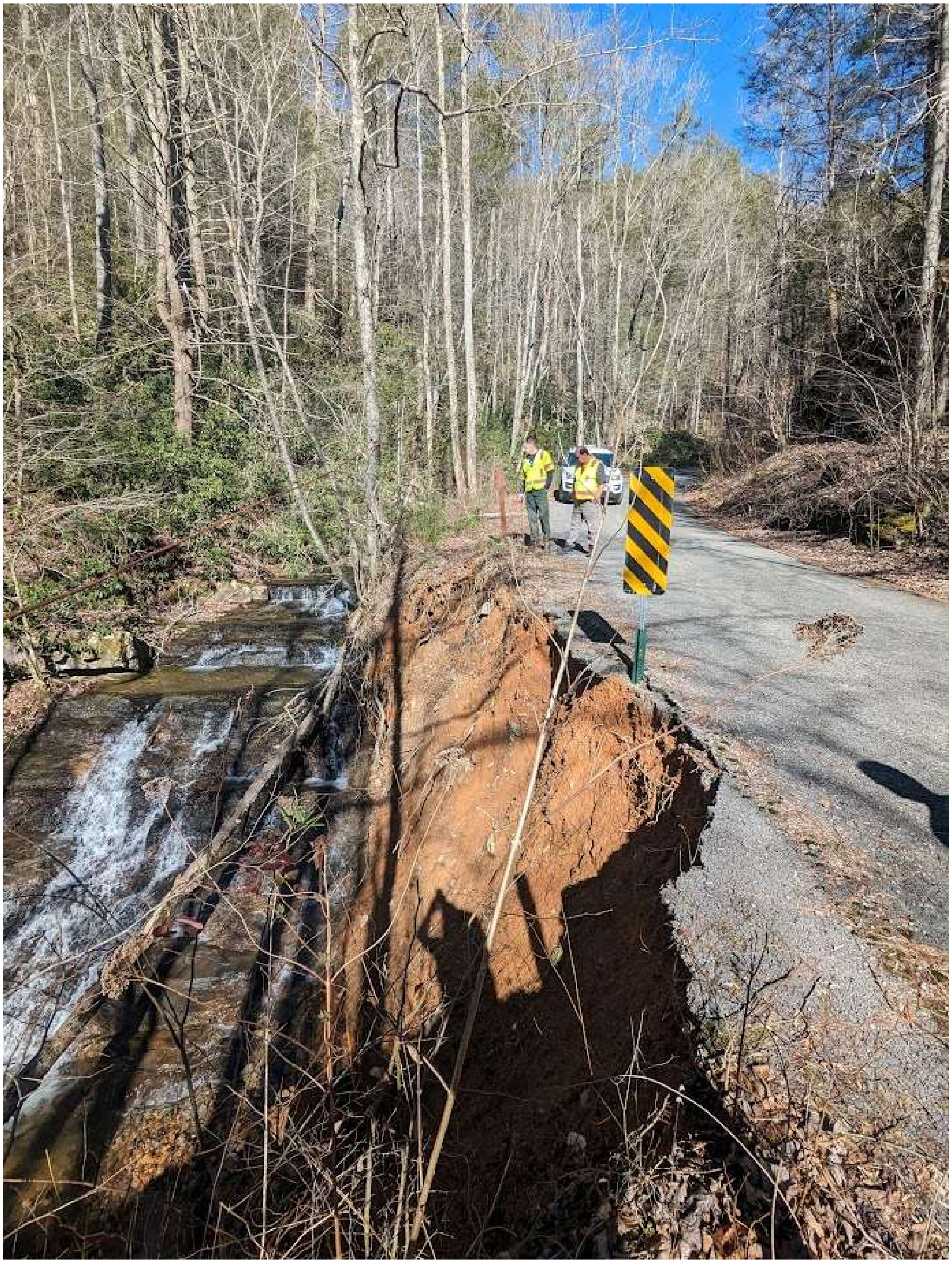
Site Photographs

Important Information About Your Geotechnical Engineering Report

Date: 2/15/2024

Photographer: Pen Sibley

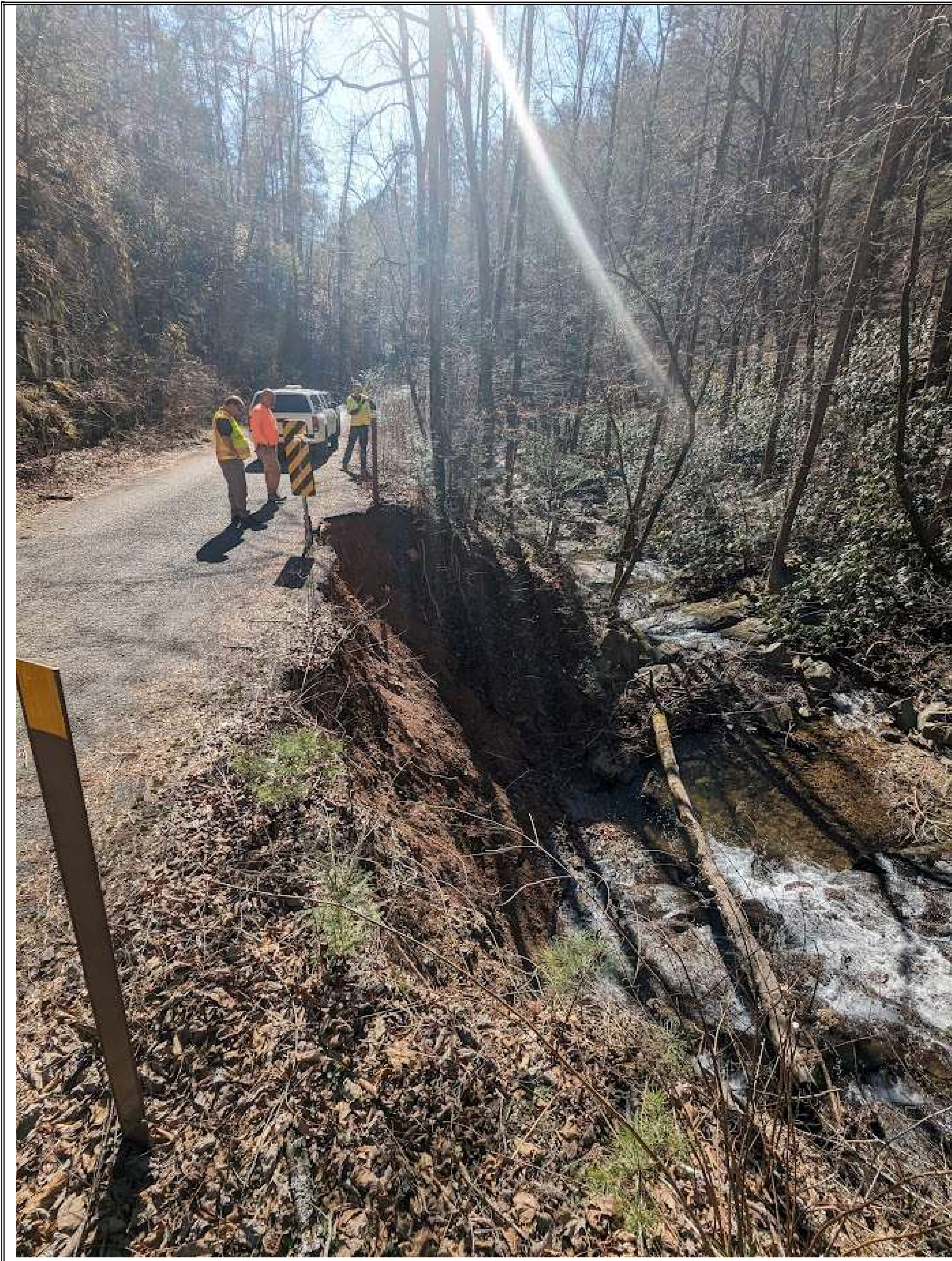
| | | |
|---|---|---|
| 1 | **Location / Orientation** | B-01 – Master Pin 77/Mile Post 6.229 |
| | **Remarks** | Site Photograph |



Date: 2/15/2024

Photographer: Pen Sibley

| | | |
|---|---|---|
| **2** | **Location / Orientation** | B-02 – Master Pin 92/Mile Post 7.84 |
| | **Remarks** | Site Photograph |



Date: 2/15/2024

Photographer: Pen Sibley

| | | |
|---|---|---|
| 3 | **Location / Orientation** | B-03 – Master Pin 92/Mile Post 7.84 |
| | **Remarks** | Site Photograph |


Date: 2/15/2024

Photographer: Pen Sibley

| | | |
|---|---|---|
| 4 | **Location / Orientation** | B-04 – Master Pin 179/Mile Post 19.302 |
| | **Remarks** | Site Photograph |

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# Important Information About Your Geotechnical Engineering Report

*Variations in subsurface conditions can be a principal cause of construction delays, cost overruns and claims. The following information is provided to assist you in understanding and managing the risk of these variations.*

## Geotechnical Findings Are Professional Opinions

Geotechnical engineers cannot specify material properties as other design engineers do. Geotechnical material properties have a far broader range on a given site than any manufactured construction material, and some geotechnical material properties may change over time because of exposure to air and water, or human activity.

Site exploration identifies subsurface conditions at the time of exploration and only at the points where subsurface tests are performed or samples obtained. Geotechnical engineers review field and laboratory data and then apply their judgment to render professional opinions about site subsurface conditions. Their recommendations rely upon these professional opinions. Variations in the vertical and lateral extent of subsurface materials may be encountered during construction that significantly impact construction schedules, methods and material volumes. While higher levels of subsurface exploration can mitigate the risk of encountering unanticipated subsurface conditions, no level of subsurface exploration can eliminate this risk.

## Scope of Geotechnical Services

Professional geotechnical engineering judgment is required to develop a geotechnical exploration scope to obtain information necessary to support design and construction. A number of unique project factors are considered in developing the scope of geotechnical services, such as the exploration objective; the location, type, size and weight of the proposed structure; proposed site grades and improvements; the construction schedule and sequence; and the site geology.

Geotechnical engineers apply their experience with construction methods, subsurface conditions and exploration methods to develop the exploration scope. The scope of each exploration is unique based on available project and site information. Incomplete project information or constraints on the scope of exploration increases the risk of variations in subsurface conditions not being identified and addressed in the geotechnical report.

## Services Are Performed for Specific Projects

Because the scope of each geotechnical exploration is unique, each geotechnical report is unique. Subsurface conditions are explored and recommendations are made for a specific project. Subsurface information and recommendations may not be adequate for other uses. Changes in a proposed structure location, foundation loads, grades, schedule, etc. may require additional geotechnical exploration, analyses, and consultation. The geotechnical engineer should be consulted to determine if additional services are required in response to changes in proposed construction, location, loads, grades, schedule, etc.

## Geo-Environmental Issues

The equipment, techniques, and personnel used to perform a geo-environmental study differ significantly from those used for a geotechnical exploration. Indications of environmental contamination may be encountered incidental to performance of a geotechnical exploration but go unrecognized. Determination of the presence, type or extent of environmental contamination is beyond the scope of a geotechnical exploration.

## Geotechnical Recommendations Are Not Final

Recommendations are developed based on the geotechnical engineer's understanding of the proposed construction and professional opinion of site subsurface conditions. Observations and tests must be performed during construction to confirm subsurface conditions exposed by construction excavations are consistent with those assumed in development of recommendations. It is advisable to retain the geotechnical engineer that performed the exploration and developed the geotechnical recommendations to conduct tests and observations during construction. This may reduce the risk that variations in subsurface conditions will not be addressed as recommended in the geotechnical report.